Haircutting Robots: The Arrival of a New Scientific Frontier

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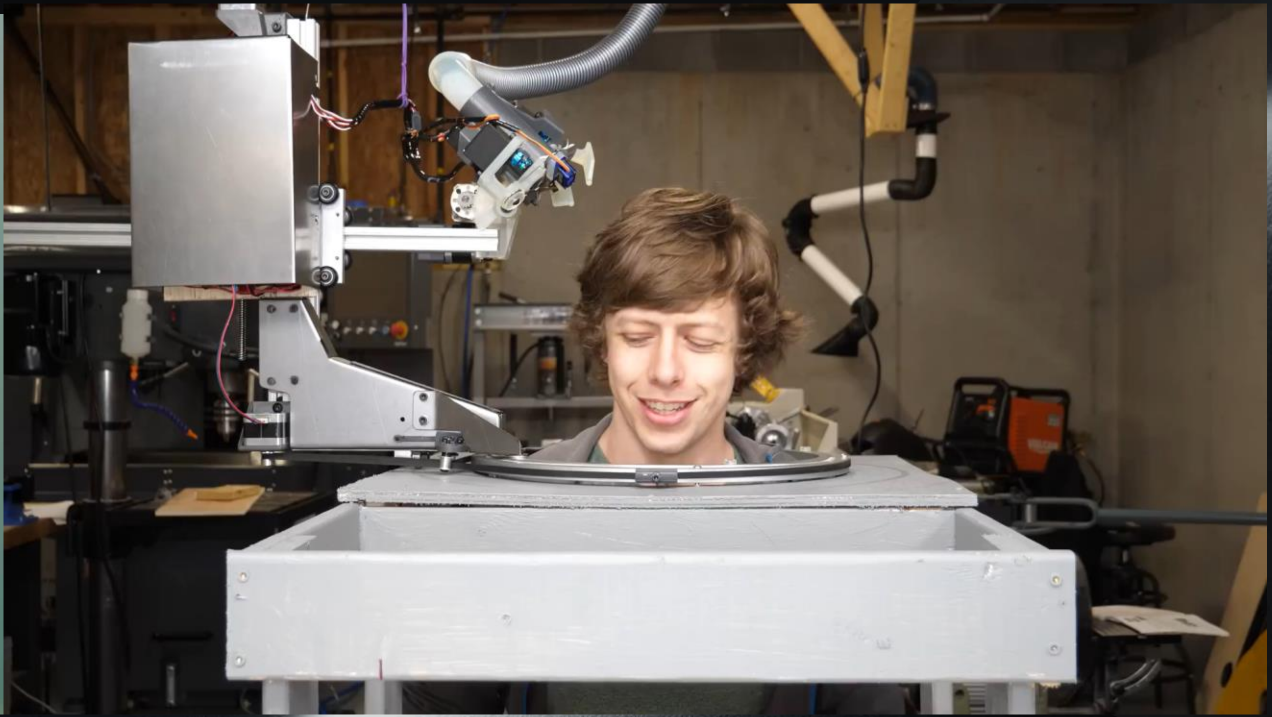
Mar. 4th 2026

CONTENTS

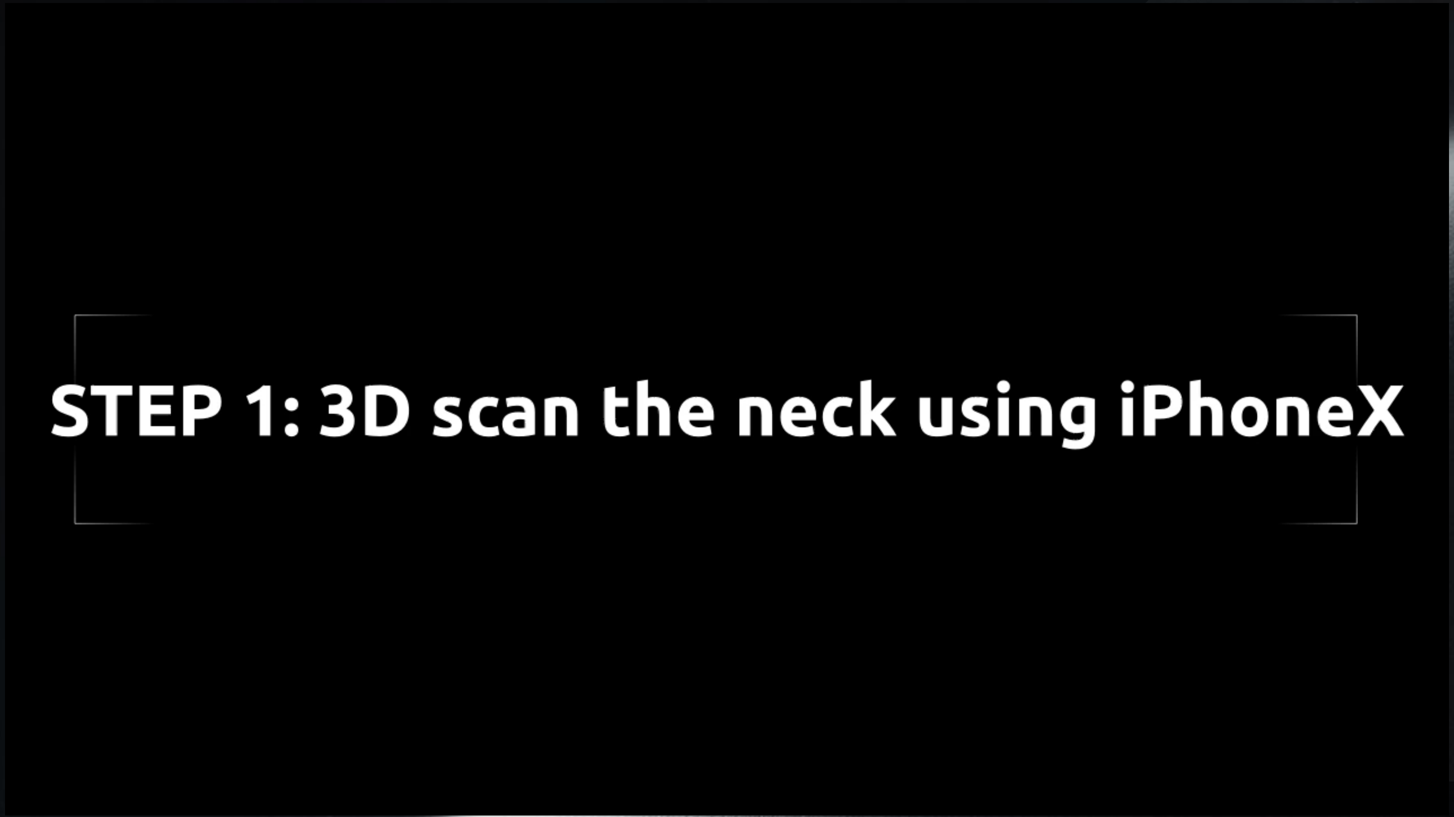
01

Introduction & Motivation

Prototypes on Haircutting Robots

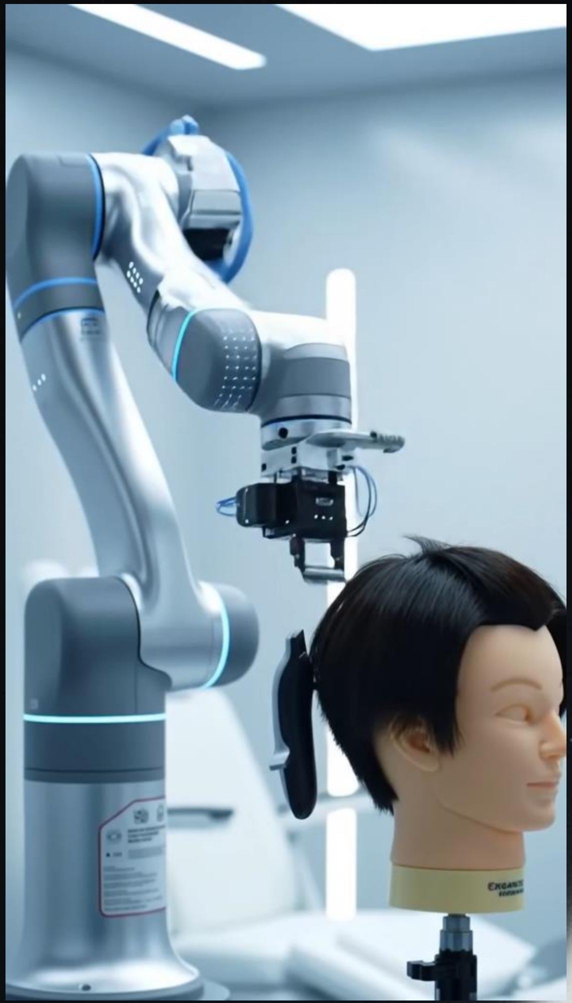


Prototypes on Haircutting Robots



Prototypes on Haircutting Robots

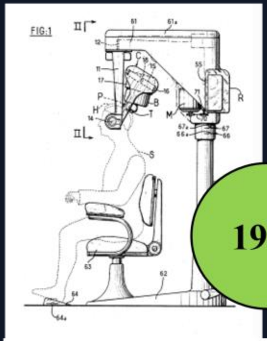
Imaginary Haircutting Robots



Imaginary Haircutting Robots

Brief History of Haircutting Robots

The first patent on haircutting




United States Patent Office

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3,241,562

AUTOMATIC HAIR-CUTTING MACHINE HAVING PROGRAMMED CONTROL MEANS FOR CUTTING HAIR IN A PREDETERMINED STYLE

Jean Gronier, 145 Blvd. de la Reine, Versailles, France

Filed Feb. 10, 1961, Ser. No. 88,409

Claims priority, application France, Feb. 16, 1960, 818,578; Feb. 1, 1961, 851,348

13 Claims. (Cl. 132—45)

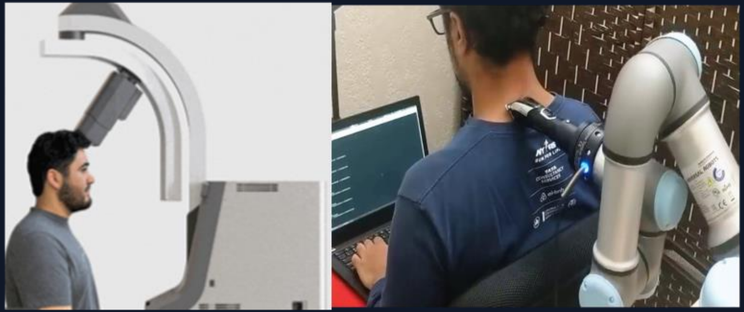
My invention has for its object a hair-cutting machine comprising means for the automatic cutting of hair in accordance with a predetermined program; each program is designed for a particular person and is established once and for all with a view to obtaining repeatedly the same cut for the same head. More specifically, the machine includes a frame, implements for cutting the hair, movable means connecting the instruments with the frame,

from above the arra... third change speed g... FIG. 19 is a pers... 18 of the same parts... FIG. 20 is a persp... the connection bet... and the casing carry... FIGS. 21, 22, 23... third change speed... ferent shafts formin... FIG. 25 is a view... the casing carrying... in the direction of t... FIG. 26 is a diag... gear system controlli... towards or away fro... FIG. 27 is a long... ment carrier.

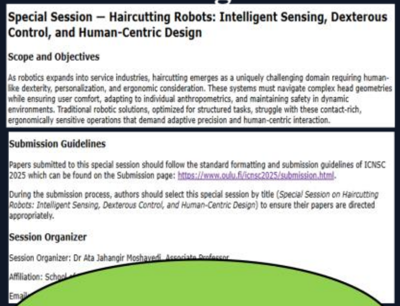
1961

Covid

A burst of DIY robotic haircutting



Conference special issue on haircutting robots


Special Session — Haircutting Robots: Intelligent Sensing, Dexterous Control, and Human-Centric Design

Scope and Objectives

As robotics expands into service industries, haircutting emerges as a uniquely challenging domain requiring human-like dexterity, personalization, and ergonomic consideration. These systems must navigate complex head geometries while ensuring user comfort, adapting to individual anthropometrics, and maintaining safety in dynamic environments. Traditional robotic solutions, optimized for structured tasks, struggle with these contact-rich, ergonomically sensitive operations that demand adaptive precision and human-centric interaction.

Submission Guidelines

Papers submitted to this special session should follow the standard formatting and submission guidelines of ICNSC 2025 which can be found on the Submission page: https://www.oulu.fi/icnsc2025/submission.html.

During the submission process, authors should select this special session by title (*Special Session on Haircutting Robots: Intelligent Sensing, Dexterous Control, and Human-Centric Design*) to ensure their papers are directed appropriately.

Session Organizer

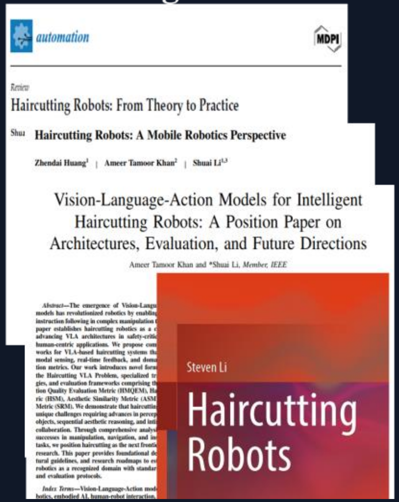
Session Organizer: Dr Ata Jahangir Moshayedi, Associate Professor

Affiliation: School...

Email...

2025

Scientific publications on haircutting robots



AI Video Generation

A burst of AI-generated videos on robotic haircutting

Keynote speech: IEEE 25th International Symposium on Computational Intelligence and Informatics (CINTI 2025) on November 18–20, 2025 in Budapest, Hungary.

Keynote speech: IEEE 2nd International Conference on Advanced Computing and Emerging Technologies (ACET 2025) on November 21, 2025.

Why Haircutting Robots Matter

Market Drivers

- Rising labor costs & barber shortages
- Demand for consistent, high-quality service
- 24/7 availability and enhanced hygiene

Robotic Solutions

- Create a new sector in automation
- Offer personalized, efficient experiences
- Address a massive, recurring market need

Rise of Affordable Robots

Then (Honda ASIMO)
~$1.3M

Now (Unitree R1)
~$5,900

This ~200x cost reduction unlocks new service sectors like haircutting, previously economically unfeasible.

02

Market Opportunity

Urban Haircut Spending

Annual spending in major cities demonstrates strong local demand for automated solutions.

NYC annual market size: 4.125 million men × 12 × 20USD × 2 = 1.98 billion USD

~2.0	~9.0	~3.5
New York City Billion USD	Shanghai Billion RMB	London Billion GBP

Global Market Estimate

Segmenting the global population reveals a massive market opportunity.

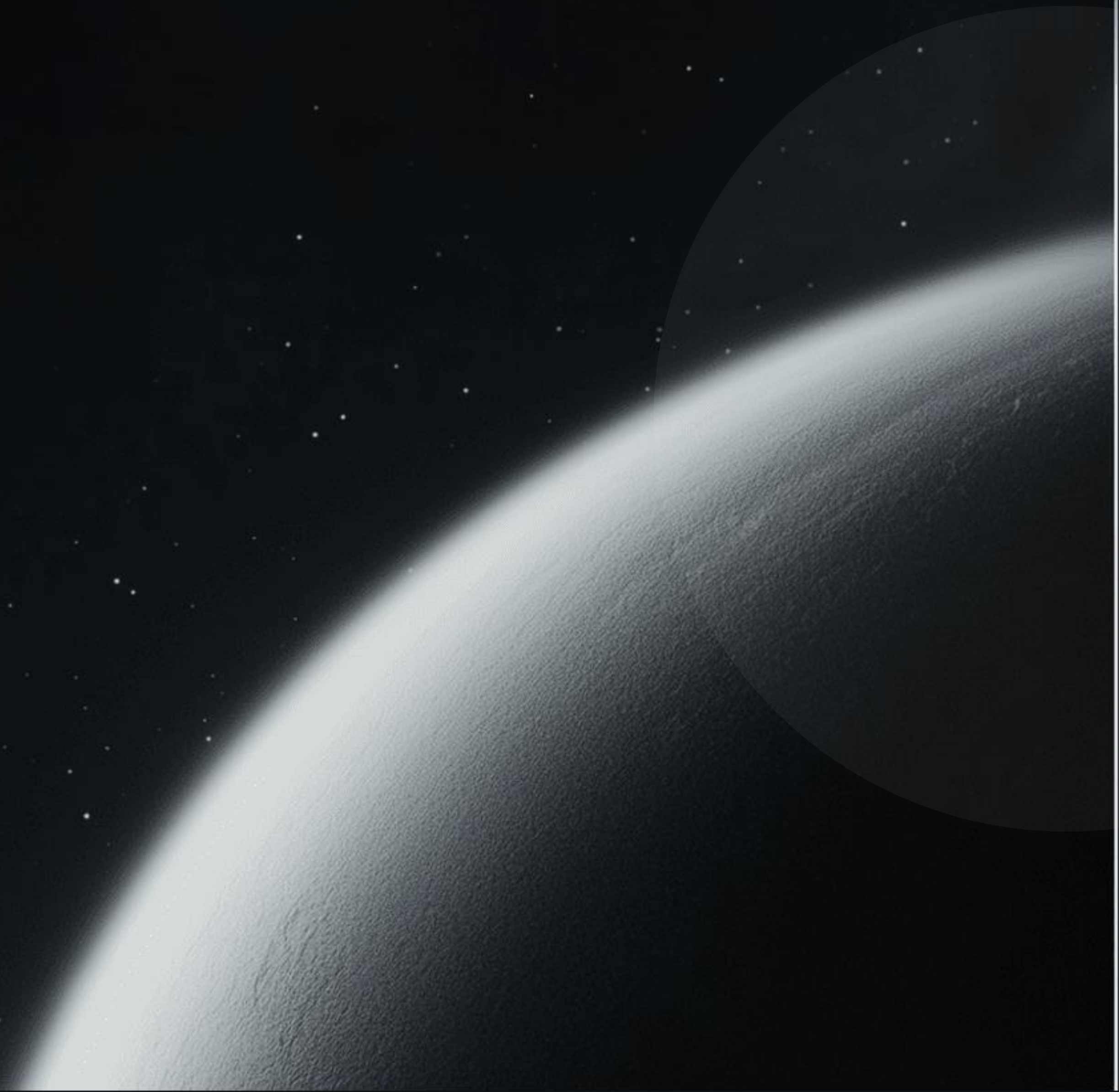
$576B High-Income 1.2B People	$216B Middle-Income 2.0B People	$35B Low-Income 1.45B People

Total Annual Global Market: ~$827 Billion USD

Ref. S. Li, Haircutting Robots: From Theory to Practice, MDPI Automation

03

Prior Research



Computer Graphics Foundations

Table 1. Existing works on hair modeling and hair editing.

References	Tasks	Application to Haircutting Robots	Code Availability
[3]	Hair simulation	Hair preview	No
[4]	Hair 3D reconstruction	2D image to hair 3D model	https://github.com/hoseok-tong/NeuralHaircutTHS
[5]	Hair segmentation	Hair preview and recommendation	No
[6]	Hair 3D reconstruction from images	2D image to hair 3D model	https://github.com/papagina/Hair-Net_DataSetGeneration
[7]	Hair 3D reconstruction from images	2D image to hair 3D model	No
[8]	CT based hair 3D reconstruction	CT based hair 3D model	https://github.com/facebookre-search/CT2Hair
[9]	Hair dynamic simulation	Hair preview	No
[11]	Image based hair style and color editing	Hair preview and recommendation	https://github.com/Zlin0530/Hair-Manip
[illegible]	Strand-based 3D hair model and hair style and color editing	Hair preview and recommendation	No
[13]	Video based dynamic image manipula-tion and editing	Hair p[illegible]nd recommendation	No

Physically-based Strand Models

Simulate hair's dynamic behavior.

Single-View 3D Reconstruction

Create accurate hair models from images.

Dynamic Simulation

Enable realistic style preview.

Robotic Hair Manipulation

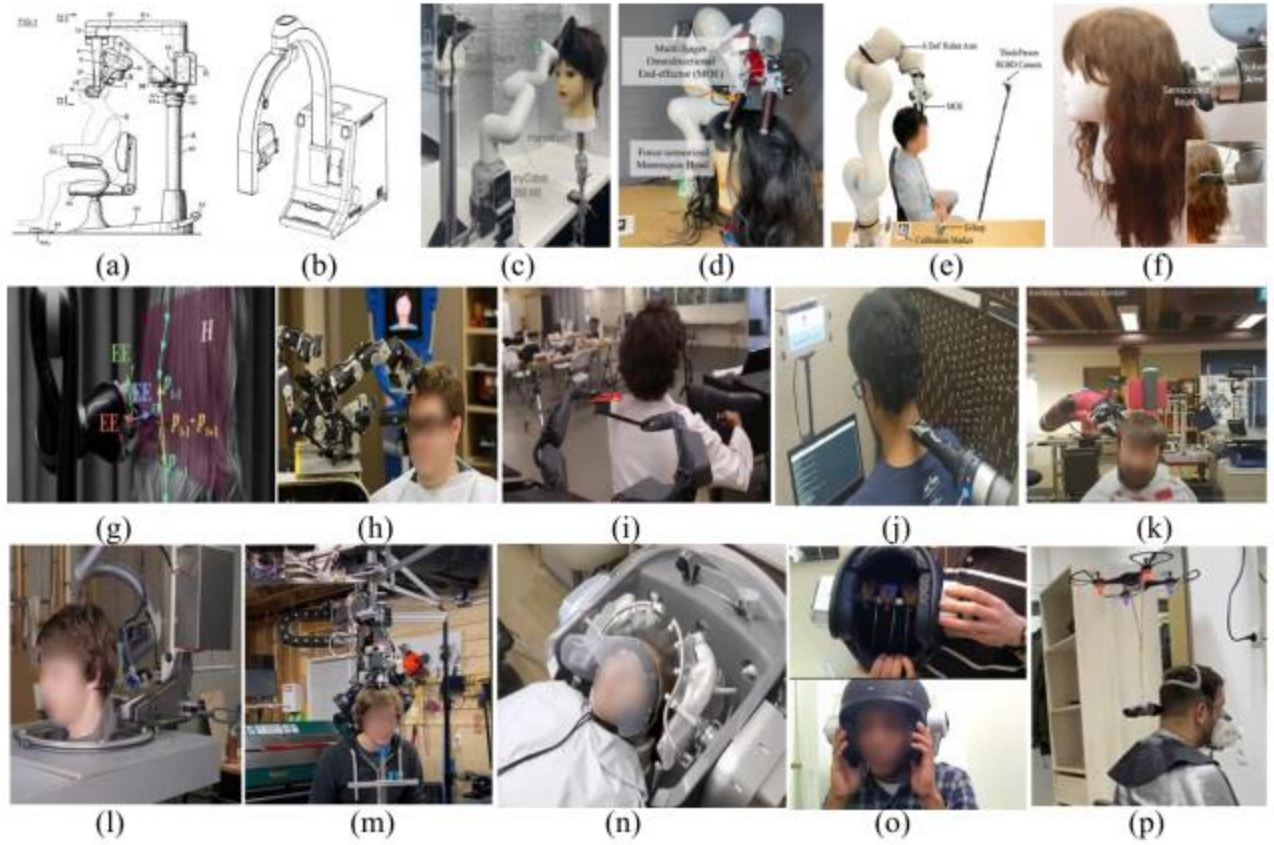


Figure. 1. Sixty-year of endeavor in human history towards haircutting robots [14,16-30]. (descriptions summarized in Table 1).

Table 2. Existing attempts towards haircutting robots.

References	No. in Fig. 1	Publication format	Type	Targeted area	Description
[14]	a	Patent	System development	Product	Mechanical and circuit design
[16]	f	Paper	Programming	Research	Coding on commercial robots with force feedback
[17]	e	Paper	Programming	Research	Coding on commercial robots with force feedback
[18]	d	Paper	System development	Research	Coding on self-developed soft gripper with commercial robots
[19]	c	Paper	Programming	Research	Coding on commercial robots for front hair styling
[20]	g	Paper	Programming	Research	Coding on commercial robots for vision-based hair combing
[21]	n	Paper	Research	Product	A commercial product by Panasonic Ltd for shampooing and scalp massaging
[22]	b	Patent	System development	Product	A commercial product by StudioRed Ltd for MyBarberRobot.
[23]	l	Video	System development	DIY	A vision-based DIY system for automated haircutting
[24]	m	Video	System development	DIY	A vision-based DIY system for automated haircutting
[25]	j	Video	Programming	DIY	iPhone scanning based automated haircutting using a commercial robot
[26]	k	Video	Programming	DIY	Automated haircutting using a collaborative robot
[27]	h	Video	Programming	DIY	Teleoperated haircutting
[28]	i	Video	Programming	DIY	Teleoperated haircutting
[29]	o	Video	Future	DIY	Imaginary haircutting helmet
[30]	p	Video	Future	DIY	Conceptual design with drones for haircutting

Refs. S. Li, Haircutting Robots: From Theory to Practice, MDPI Automation;
S. Li, Book: Haircutting Robots, Springer.

04

Technical Requirements

Teleoperated Cutting Needs

Remote operation must overcome fundamental communication delays to ensure stability and safety.

Figure 2. Teleoperated haircutting, which allows remote haircut of customers in metropolitan areas by barbers living in rural areas. (This figure is partially edited with Gemini 2.5 Flash).

Ref. S. Li, Haircutting Robots: From Theory to Practice, MDPI Automation

Autonomous Haircutting

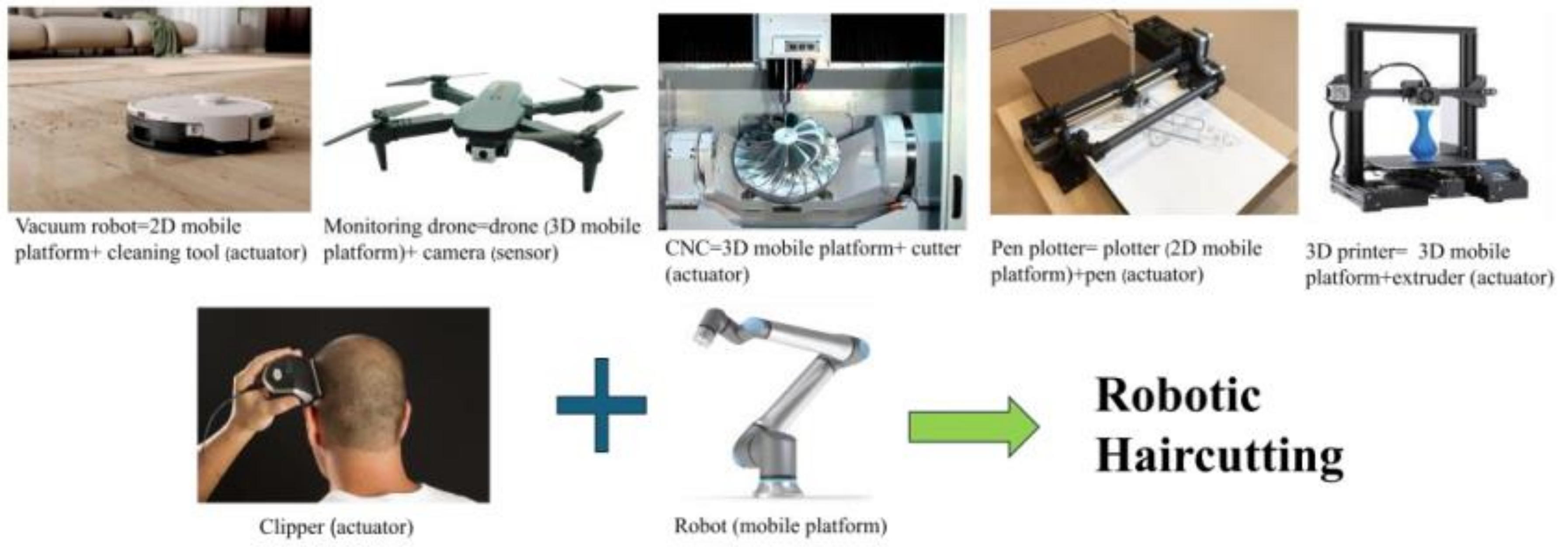
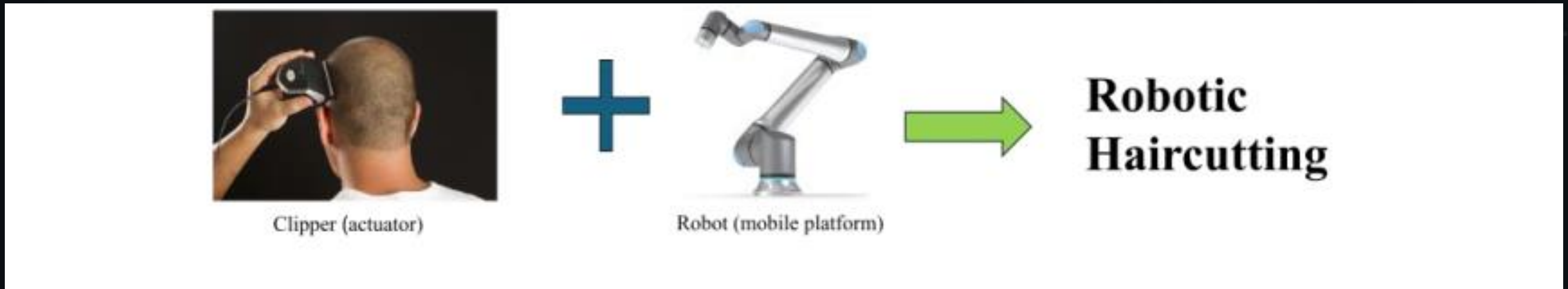


Figure 3. Many robotic systems can be viewed as the combination of mobile platforms and sensors/actuators. In this perspective, a haircutting robot can be viewed as a clipper as an actuator plus a robotic arm as the mobile platform.

Refs. S. Li, Haircutting Robots: From Theory to Practice, MDPI Automation
A. Khan and S. Li Robotic Haircutting Systems: A Survey of Methods, Challenges, and Hair Modeling Insights, IEEE Journal of Selected Areas in Sensors.

Autonomous Cutting Pipeline



UFO Clipper

Flowbee Clipper

Autonomous Cutting Pipeline: A CNC Perspective

1. Workpiece Definition
Treat head as a dynamic 3D object.

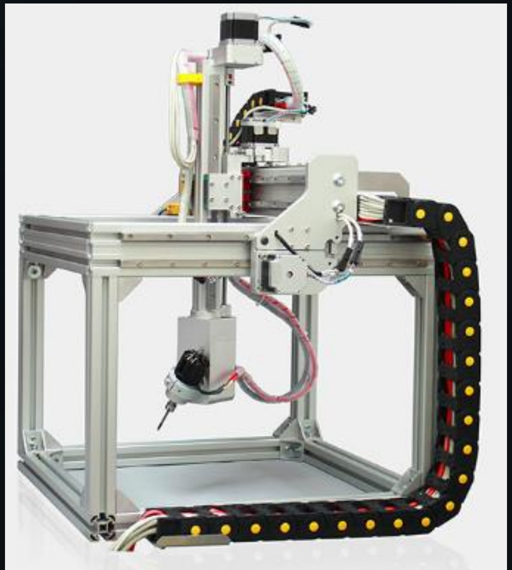
2. 5-Axis Toolpath Planning
Generate precise cutting trajectories.

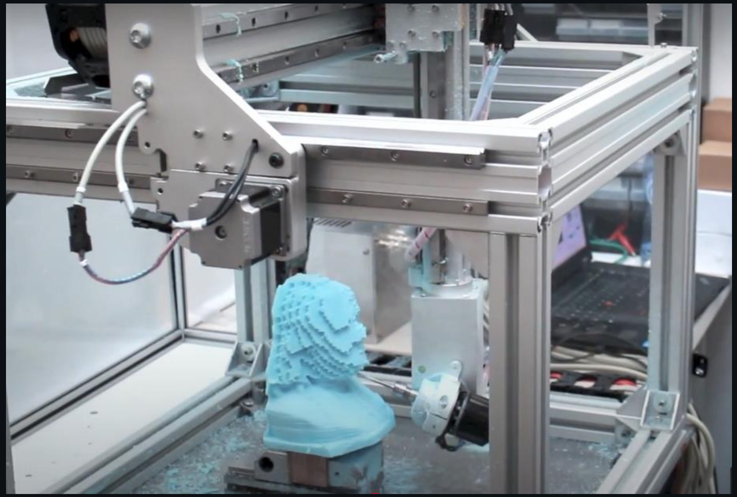
3. Real-Time Tracking
Adapt to head motion with visual servoing.

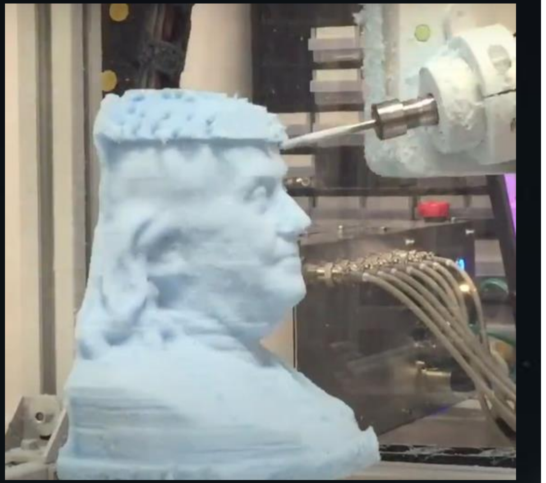
CNC for Sculpture

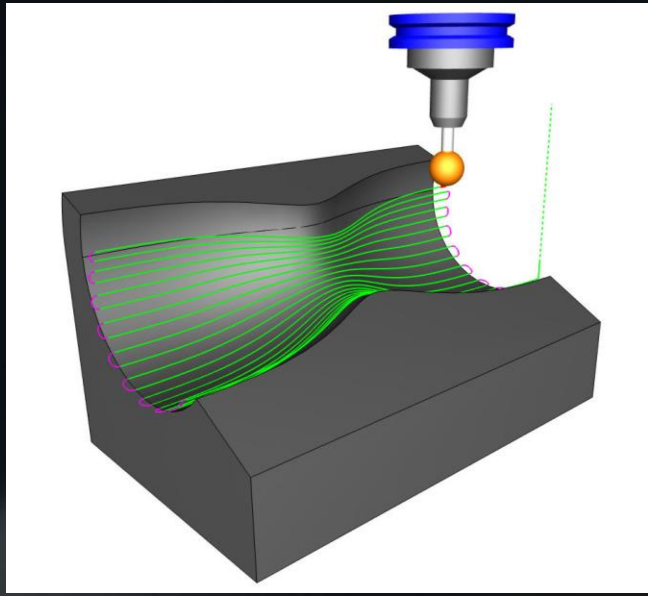
KUKA robot for wood carving

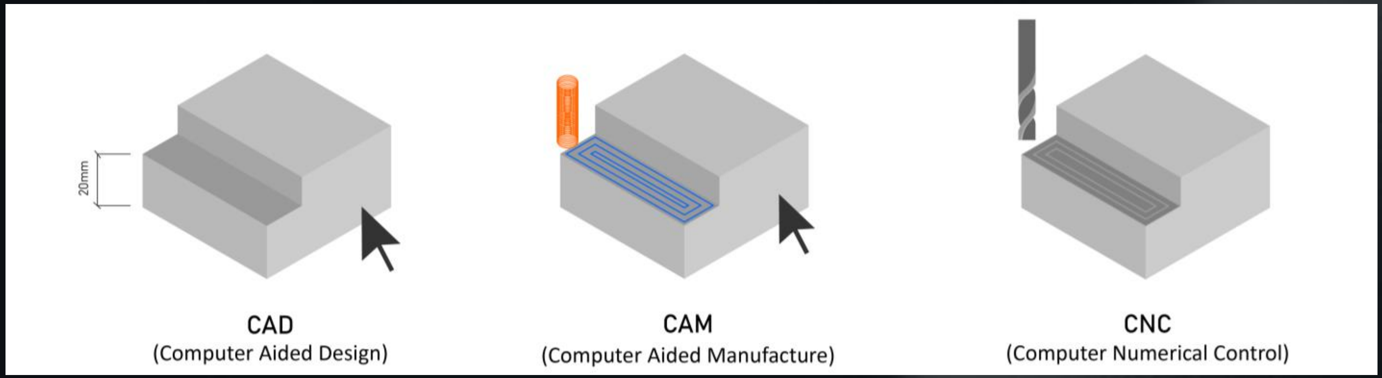
CNC Perspective





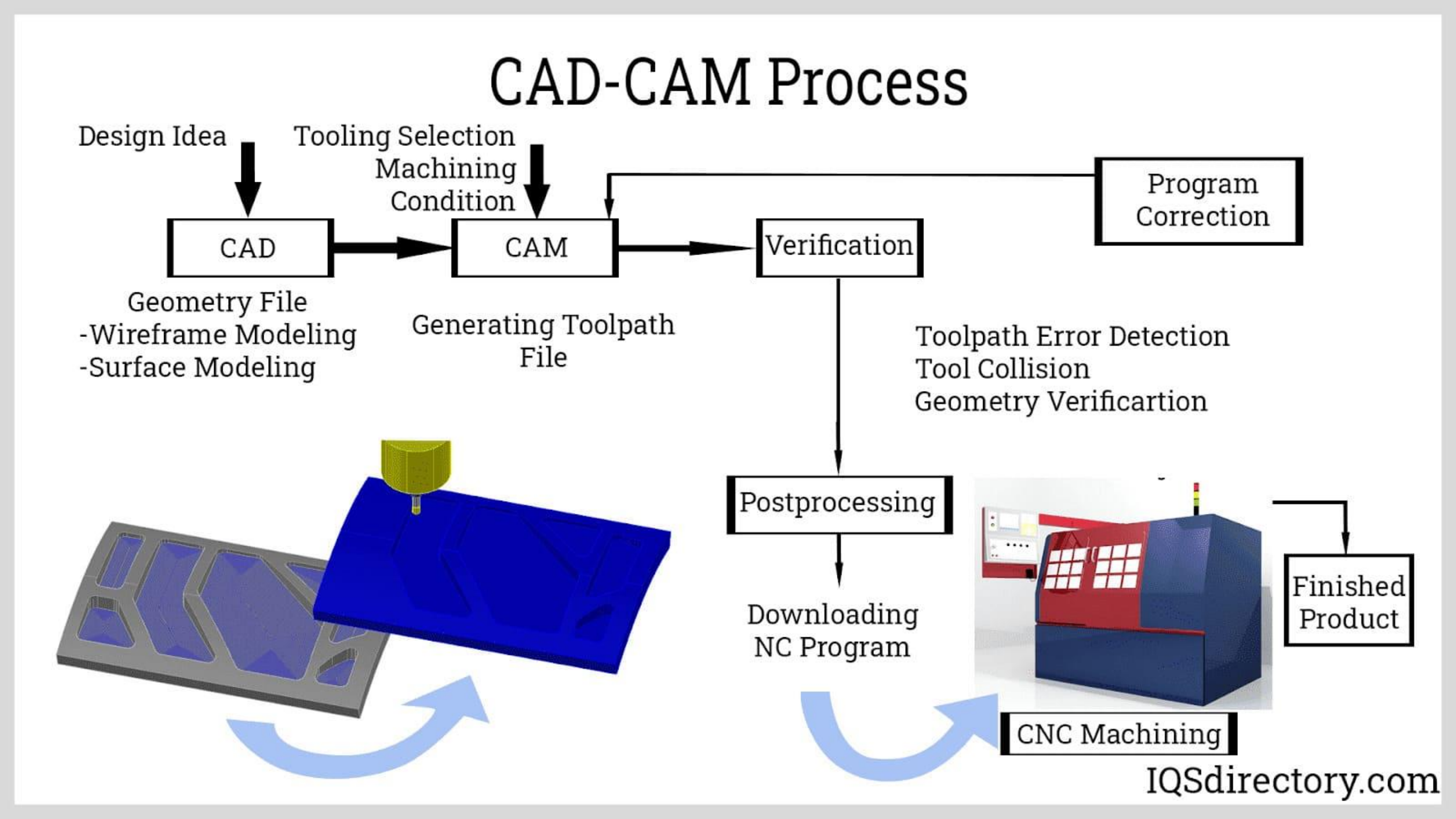






Ref: A. Khan and S. Li, CNC-Inspired Robotic Hair Cutting: A Comprehensive Survey on Precision Personal Care Automation, Journal of Artificial Intelligence for Automation

CNC Perspective



https://www.iqsdirectory.com/articles/cnc-machining.html

Ref: A. Khan and S. Li, CNC-Inspired Robotic Hair Cutting: A Comprehensive Survey on Precision Personal Care Automation, Journal of Artificial Intelligence for Automation

CNC Perspective

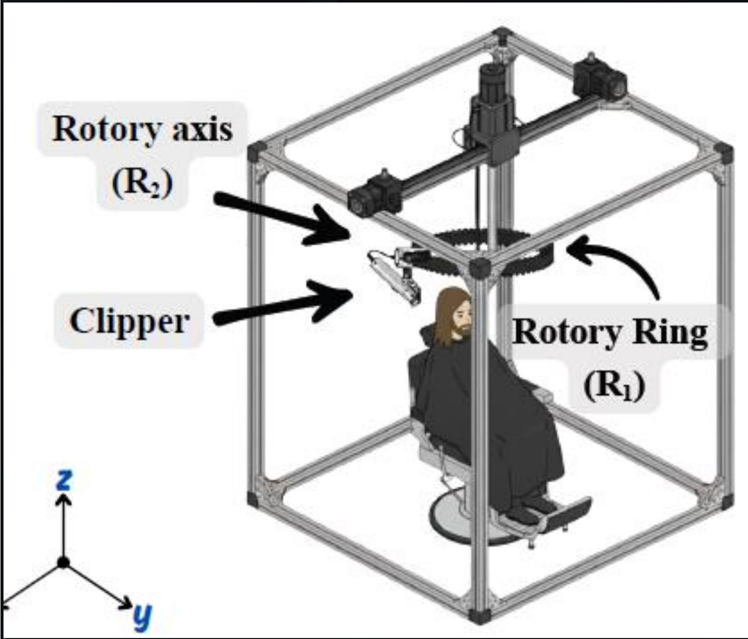


Figure 3. Toolpath planning over a 3D head model showing contour-parallel trajectories, tool orientation vectors, and a zoomed inset highlighting local path detail for controlled surface-following hair-cutting motion.

Algorithm 1 Multi-Pass Toolpath Generation

Require: Initial hair volume S_0, Target hairstyle T, Safety zones O
Ensure: Sequence of toolpaths $\{g_i(t)\}_{i=1}^{N}$

Initialize residual volume $S \leftarrow S_0$
$i \leftarrow 0$
while $S \not\subseteq T^{+\epsilon}$ **do**
  $i \leftarrow i + 1$
  Compute removal target $R_i \leftarrow S \setminus T$
  Generate collision-free toolpath $g_i(t)$ covering R_i
  Verify $g_i(t) \cap O = \emptyset$ for all t
  Execute pass and update $S \leftarrow S \setminus E_i$
end while
return $\{g_i(t)\}_{i=1}^{N}$

Ref: A. Khan and S. Li, CNC-Inspired Robotic Hair Cutting: A Comprehensive Survey on Precision Personal Care Automation, Journal of Artificial Intelligence for Automation

Mobile Robotics Perspective

FIGURE 1 Analogy between robotic lawn mowing and robotic haircutting, highlighting the natural extension of mobile robot techniques from grass mowing to hair trimming.

Continuous data collection creates a virtuous cycle of improvement.

Haircut
Motion & Sensor Data

→

Feedback
Ratings & Reviews

→

Training
ML Model Improvement

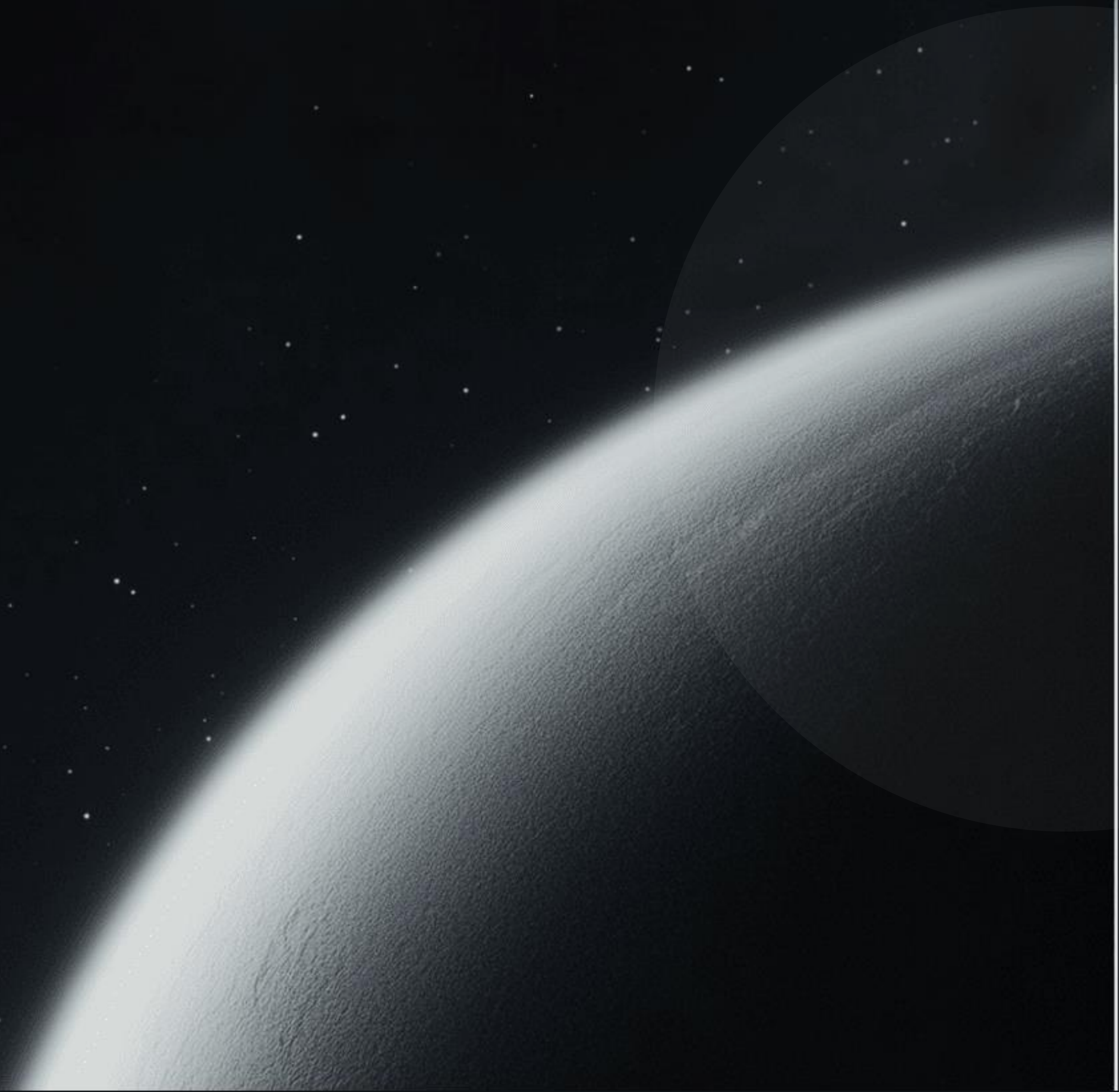
→

Better Service
Enhanced Accuracy

Ref. Z. Huang, A. Khan and S. Li, Haircutting Robots: A Mobile Robotics, Submitted.

05

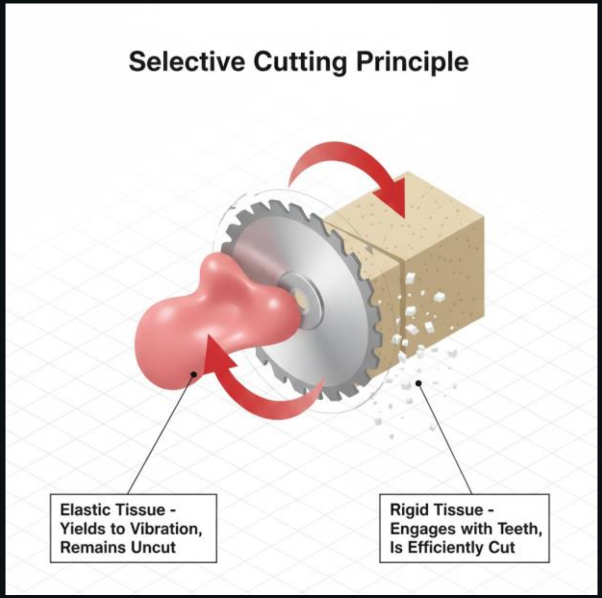
Safety & Security



Safety Certification

Ensuring client safety is the top priority, requiring a multi-layered approach to mitigate risks.

Sharp Tools: Use safer alternatives like clippers.



Safety Certification

Ensuring client safety is the top priority, requiring a multi-layered approach to mitigate risks.

TABLE 2 Categorization of risks in haircutting robots across system aspects

Risk category	Typical issues	Associated components
Network & cybersecurity	• High latency in teleoperation causing delayed reaction • Network congestion leading to packet loss • Cyberattacks that hijack robot control • Unauthorized modification of haircut trajectory • Service interruption from unstable wireless connection	Communication link, wireless modules, teleoperation interface, network protocols
Perception & sensing	• Misclassification of hair, skin, or ear boundaries • Depth camera errors due to lighting or occlusion • Force/torque sensor drift or calibration failure • Loss of vibration or acoustic signals during tool–hair interaction • Single-sensor dependency without redundancy	Cameras, depth sensors, force/torque sensors, acoustic/vibration sensors, sensor fusion modules
Physical & tool-related	• Excessive contact force from clippers or scissors • Unsafe blade geometry causing skin injury • Overheating of dryers or curlers near scalp • Accidental collision with robot arm or end-effector • Tool wear leading to irregular performance	Clippers, scissors, heating tools (dryer/curler), blade geometry, robotic arm end-effector
Control & algorithmic	• Instability of controller under time delay • Unsafe motion planning without safety margins • Lack of fallback strategies when sensors fail • Runtime monitoring failure to stop hazardous action • Inconsistent tool switching logic	Motion planner, obstacle avoidance module, controller, runtime safety monitor, tool-switching logic
Human-related	• Sudden head movements (rotation, tilting, reflexes) • Postural shifts due to discomfort or fatigue • Fear or psychological stress during close contact • Non-cooperation or unexpected user actions • Lack of trust or acceptance of robotic haircutting	User head, posture, facial regions (ear/neck), behavioral response, psychological state

Sharp Tools: Use safer alternatives like clippers.

Excessive Force: Implement soft, compliant materials.

Sensor Failure: Use redundant sensor systems.

Cybersecurity: Employ local safety monitors.

Z. Huang, A. Vilkki, S. Li and J. Röning, Safety in Robotic Haircutting, Under 2nd round of review in Journal of Field Robotics

Safety Certification

Ensuring client safety is the top priority, requiring a multi-layered approach to mitigate risks.

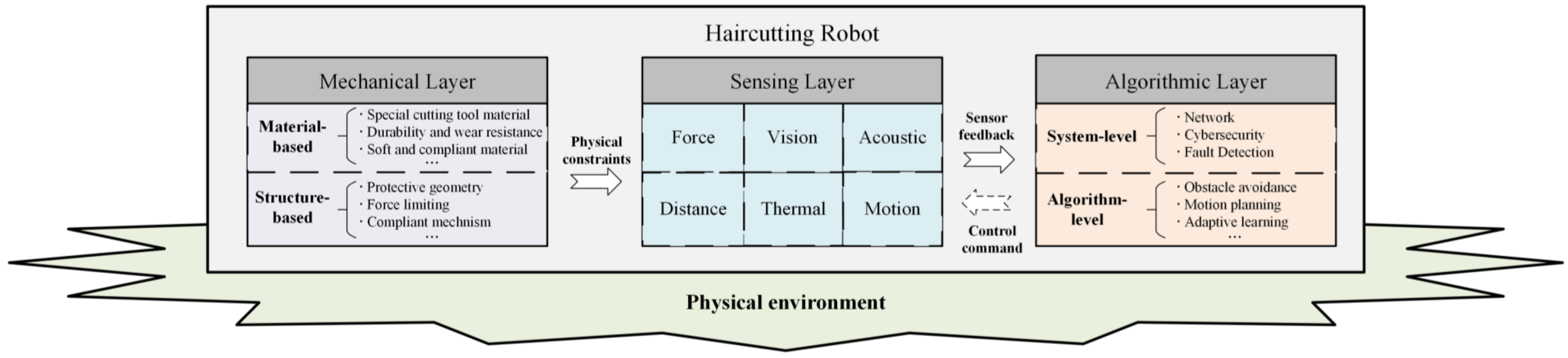


FIGURE 3 Three-layer safety framework for haircutting robots with layer-wise components

- Sharp Tools: Use safer alternatives like clippers.
- Sensor Failure: Use redundant sensor systems.
- Excessive Force: Implement soft, compliant materials.
- Cybersecurity: Employ local safety monitors.

Safety Certification

Ensuring client safety is the top priority, requiring a multi-layered approach to mitigate risks.

Supervised Autonomy Model

Cybersecurity: Employ local safety monitors.

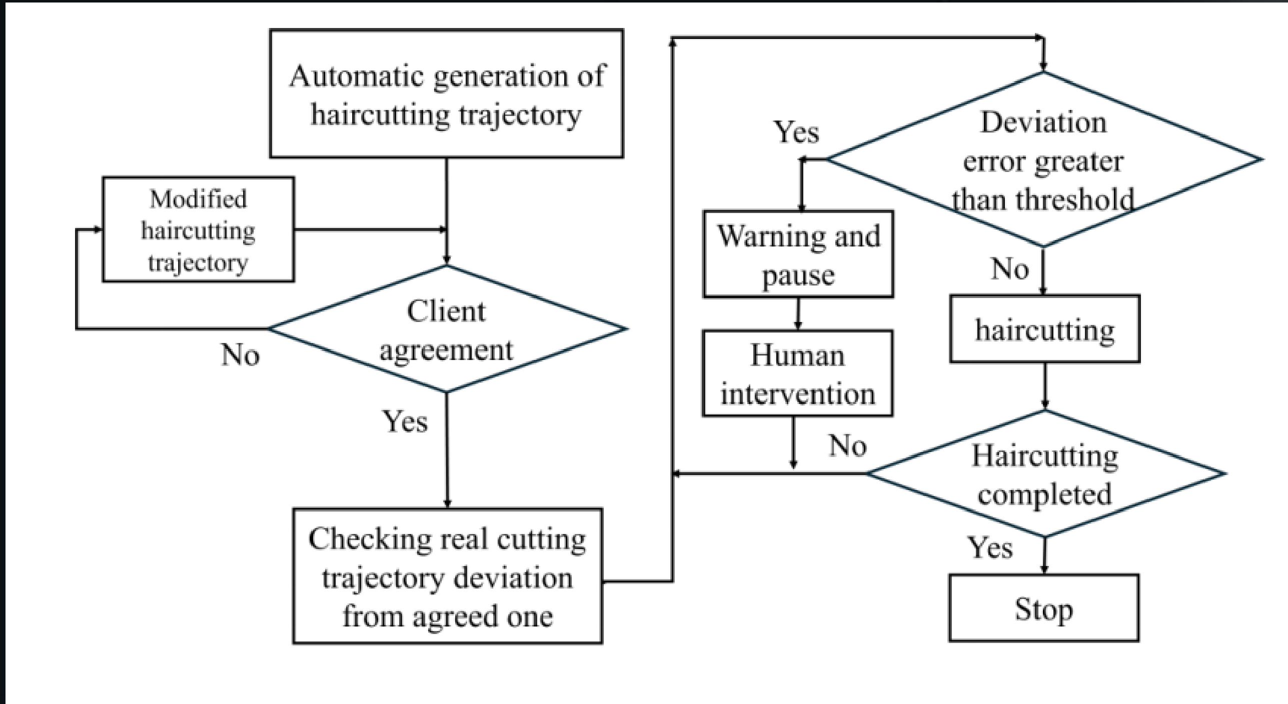


Figure 4. Flowchart of robotic haircutting process with possible human intervention.

Intervention is triggered by deviations, safety alerts, obstacles, or requests.

Supervised Autonomy Model

Full autonomy is deferred in favor of a practical model where one barber oversees multiple robots.

06

Empowering Technologies

Empowering Tech: Drive & AI

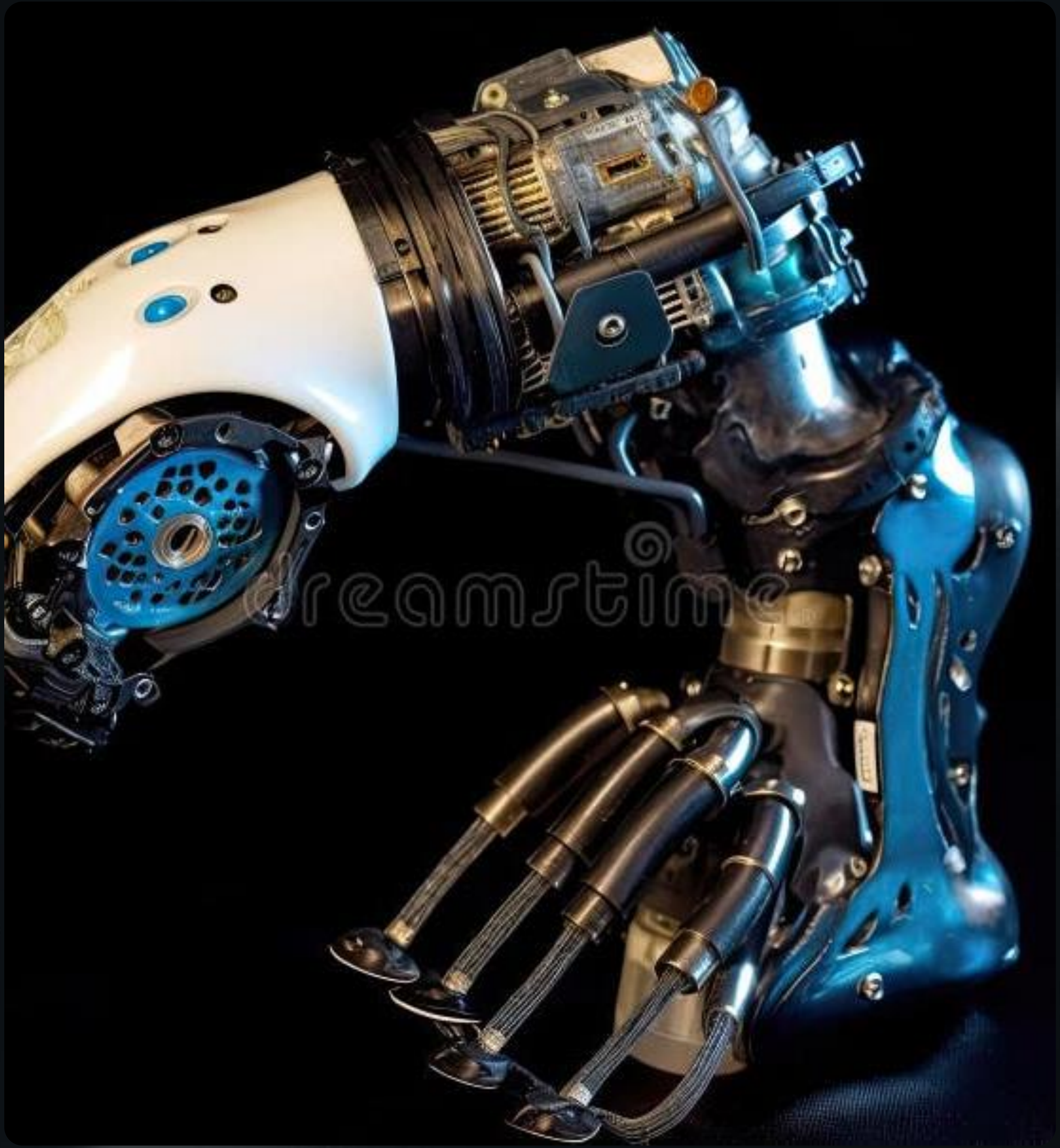
Direct Drive Technology

Provides precise, backlash-free motion crucial for intricate cuts and real-time head movement compensation.

Large Language Models (AI)

Offer conversational companionship and personalized style advice, enhancing the customer experience.

Ref. S. Li, Book: Haircutting Robots, Springer.



Empowering Tech: Immersion & Touch

Virtual Reality (VR)

Immerses remote barbers in the client's environment for precise assessment and control.

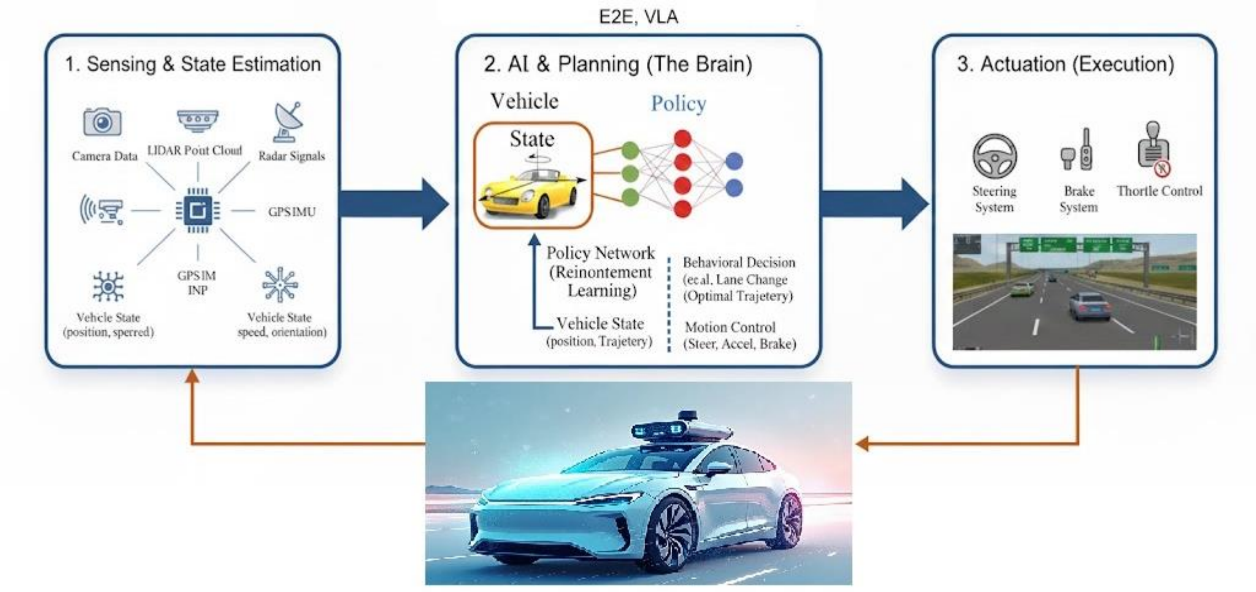
Force Control and Haptic Feedback

Allows operators to feel cutting forces, enabling fine control and a safer experience.

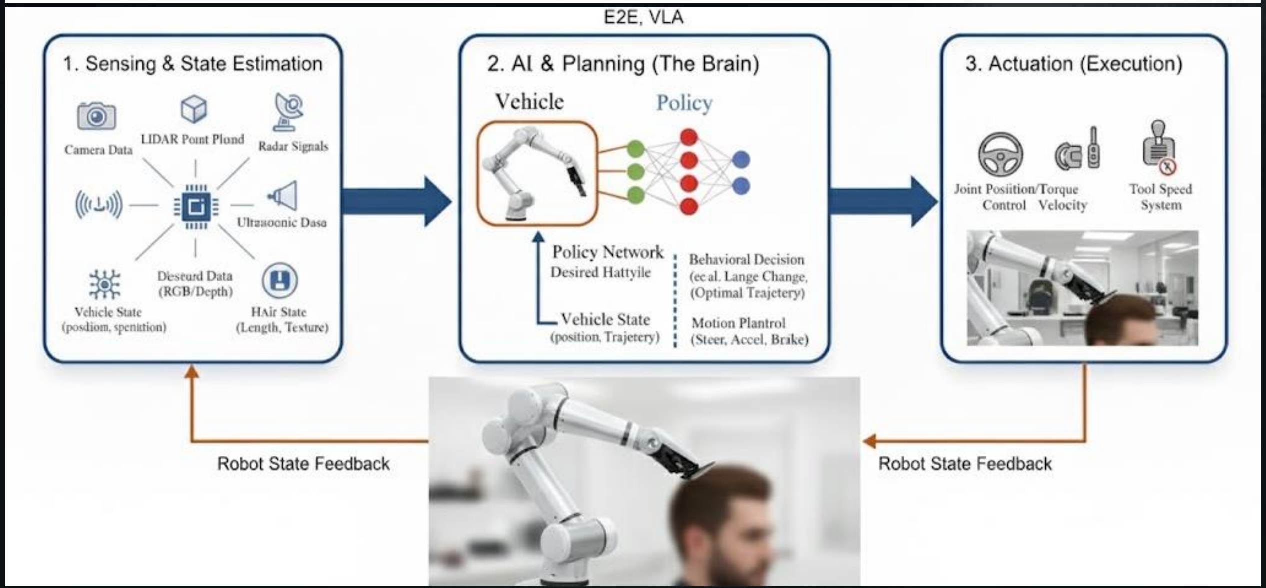
Ref. S. Li, Book: Haircutting Robots, Springer.

Empowering Tech: E2E, VLA, Imitation Learning

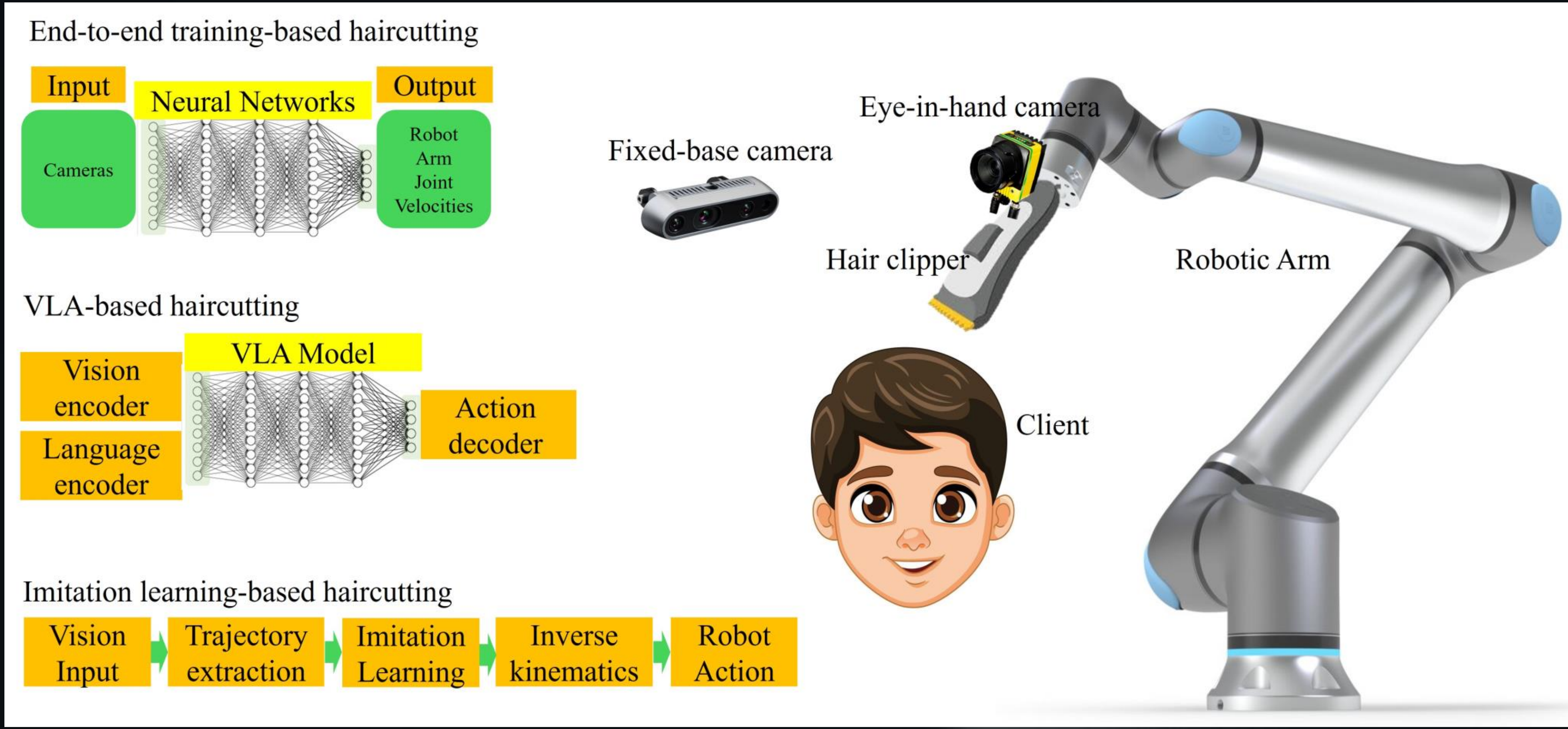
Autonomous Driving



Autonomous Haircutting

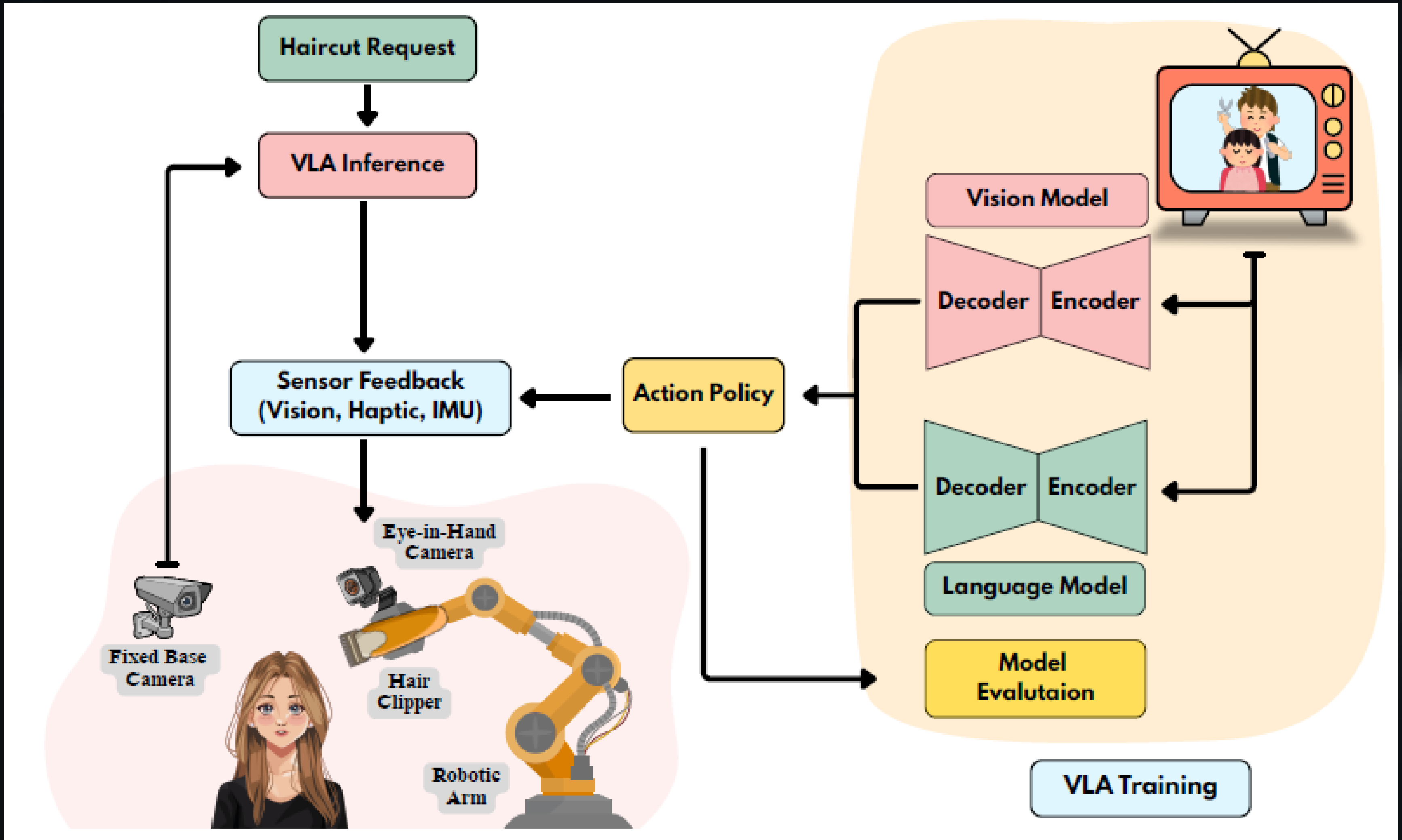


Empowering Tech: E2E, VLA, Imitation



Ref: A. Khan and S. Li, Vision-Language-Action Models for Intelligent Haircutting Robots: A Position Paper on Architectures, Evaluation, and Future Directions, Submitted.

Empowering Tech: E2E, VLA, Imitation



Ref: A. Khan and S. Li, Vision-Language-Action Models for Intelligent Haircutting Robots: A Position Paper on Architectures, Evaluation, and Future Directions, Submitted.

Core Supporting Technology for Haircutting Robots

Perception Layer

- 3D Visual Modeling (Structured Light)
- Force/Tactile Sensing (Anti-scratch)
- Hair Feature Recognition (Hardness/Density)

Decision & Algorithm

- AI Hairstyle Design (Face Shape Matching)
- Motion Path Planning (Obstacle Avoidance)
- Intelligent Control (PID, Deep Learning)

Execution Layer

- Multi-DOF Robotic Arm (6/7-axis)
- End Effector (Electric Clipper, Scissors)
- Direct Drive Servo (High Precision)

HCI Layer

- Voice/Gesture Control Interaction
- AR Hairstyle Preview & Try-on
- Remote Control & Real-time Monitoring

Safety & Reliability

- Collision Emergency Stop & Force Limit
- Power/Network Failure Safe Return
- Fault Detection & Predictive Maintenance

Testing & Industrialization

- Function/Performance/Safety Testing
- Modular Design & Mass Production
- Business Model & Industry Standards

This six-layer closed-loop system integrates interdisciplinary technologies, enabling fully automated and safe haircut services.

I. Perception Layer: Environment & Human Understanding

Core Objective: Achieve precise recognition and understanding of the head, hairstyle, and environment.

3D Visual Modeling

Tech: Structured Light, ToF, LiDAR
Func: Reconstruct head model, extract hair contours/lengths for path planning.
Challenges: Real-time dynamic modeling, lighting robustness.

Force & Tactile Sensing

Tech:6-axis Force Sensors, Tactile Arrays
Func: Detect contact force, enable feedback control for safety & comfort.
Challenges: Sensor durability, real-time signal processing.

Hair Feature Perception

Tech: High-speed Cameras, Spectral Analysis
Func: Identify hardness, curliness, density for personalized cutting.
Challenges: Non-contact, fast & accurate analysis.

Summary: The Perception Layer acts as the robot's 'eyes' and 'skin'. It integrates multi-sensor data to create a precise digital twin of the user's head, enabling the system to make intelligent, safe, and personalized decisions during the haircut process.

II. Decision & Algorithm Layer: The Intelligent Brain

Core Objective: Intelligently plan and control the entire haircut process based on perception data.

AI Hairstyle Design

Tech: CV face analysis, golden ratio models, hairstyle DB.
Func: Recommends style via AR/VR preview; generates digital target model.
Chall: Quantifying aesthetics, satisfying personalized needs.

Motion & Path Planning

Tech: Inverse kinematics, RRT*, trajectory optimization.
Func: Plans collision-free paths for arm to reach target positions accurately.
Chall: High-redundancy path optimization, dynamic obstacle avoidance.

Intelligent Control

Tech: PID, adaptive control, sliding mode control, RL.
Func: Precisely controls force/speed; learns to optimize cutting strategies.
Chall: High-dynamic real-time response, robustness in complexity.

This layer acts as the intellectual core, translating visual data into actionable, precise robotic movements through advanced computational intelligence.

III. Execution Layer: Mechanics & Haircut Operation

Core Objective: Precisely and stably execute haircutting actions

Mechanical Body Design

Tech:6/7-axis robotic arms, lightweight materials, high-precision reducers.
Func: Full head coverage, obstacle avoidance, compliant operation.
Chall: Balancing precision, payload, and compactness.

End Effector (Tool)

Tech: Integrated clippers, scissors, combs with auto-adjustment.
Func: Trimming, thinning, styling; anti-hair-pinching safety.
Chall: Miniaturization, reliability, and quick tool change.

Drive System

Tech: High-precision servo motors, direct-drive, harmonic drives.
Func: Power movement, ensure precise control & fast response.
Chall: System efficiency, low noise, and long-term stability.

Translating decisions into precise physical movements with industrial-grade accuracy.

Human-Computer Interaction Layer

Core Objective: Achieve a convenient, intuitive, and user-friendly interaction experience.

Interaction Methods

Technologies: Speech recognition, computer vision gesture recognition, HD touch screens.
Function: Control robot via voice, gestures or touch. NLP enables fluent, humanized interaction.
Challenges: Accuracy in complex environments, gesture robustness, logic simplicity.

Visual Interaction

Technologies:AR/VR, 3D rendering engines.
Function: Visualize virtual hairstyles on user's head pre-cut. Display real-time progress.
Challenges: Accurate virtual-real matching, real-time rendering efficiency.

Remote Interaction

Technologies: IoT, 5G communication, remote control software.
Function: Remote booking, status monitoring, and expert remote guidance for complex styles.
Challenges: Low-latency control, network security & privacy.

The Human-Computer Interaction Layer bridges the gap between user intent and robotic execution. By integrating advanced perception and visualization technologies, we aim to make the haircutting process not just automated, but truly intelligent and empathetic.

V. Safety & Reliability Layer: Bottom-line Guarantee

Core Objective: Ensure absolute safety during the haircut and stable and reliable system operation.

Safety Mechanisms

Technologies: Emergency stop, collision detection, force feedback, safety light curtains.
Function: Multi-redundancy system. Immediate stop on abnormal force or collision. Electrical safety design.
Challenges: Fast response, avoiding false triggers.

Reliability Design

Technologies: Fault diagnosis, sensor redundancy, modular design, thermal management.
Function: Real-time monitoring, failure prediction, proactive maintenance, high availability.
Challenges: Complex system failure analysis, predictive accuracy.

Compliance & Ethics

Technologies: Data encryption, access control, privacy protocols.
Function: Strict protection of facial/hairstyle data. Compliance with safety standards and norms.
Challenges: Balancing data utilization with privacy, adapting to regulations.

This layer serves as the foundation for user trust, ensuring that innovation never compromises on safety or privacy.

VI. Testing & Industrialization Layer: Commercialization Support

Core Objective: Drive technology from the laboratory to the market and achieve large-scale application.

Testing & Validation

Technologies: Simulation platforms, HIL testing, UX testing, reliability testing.
Function: Comprehensive pre-launch tests for safety and quality. Optimize results via real-user data.
Challenges: Scenario coverage, quantitative evaluation, rapid iteration.

Industrialization System

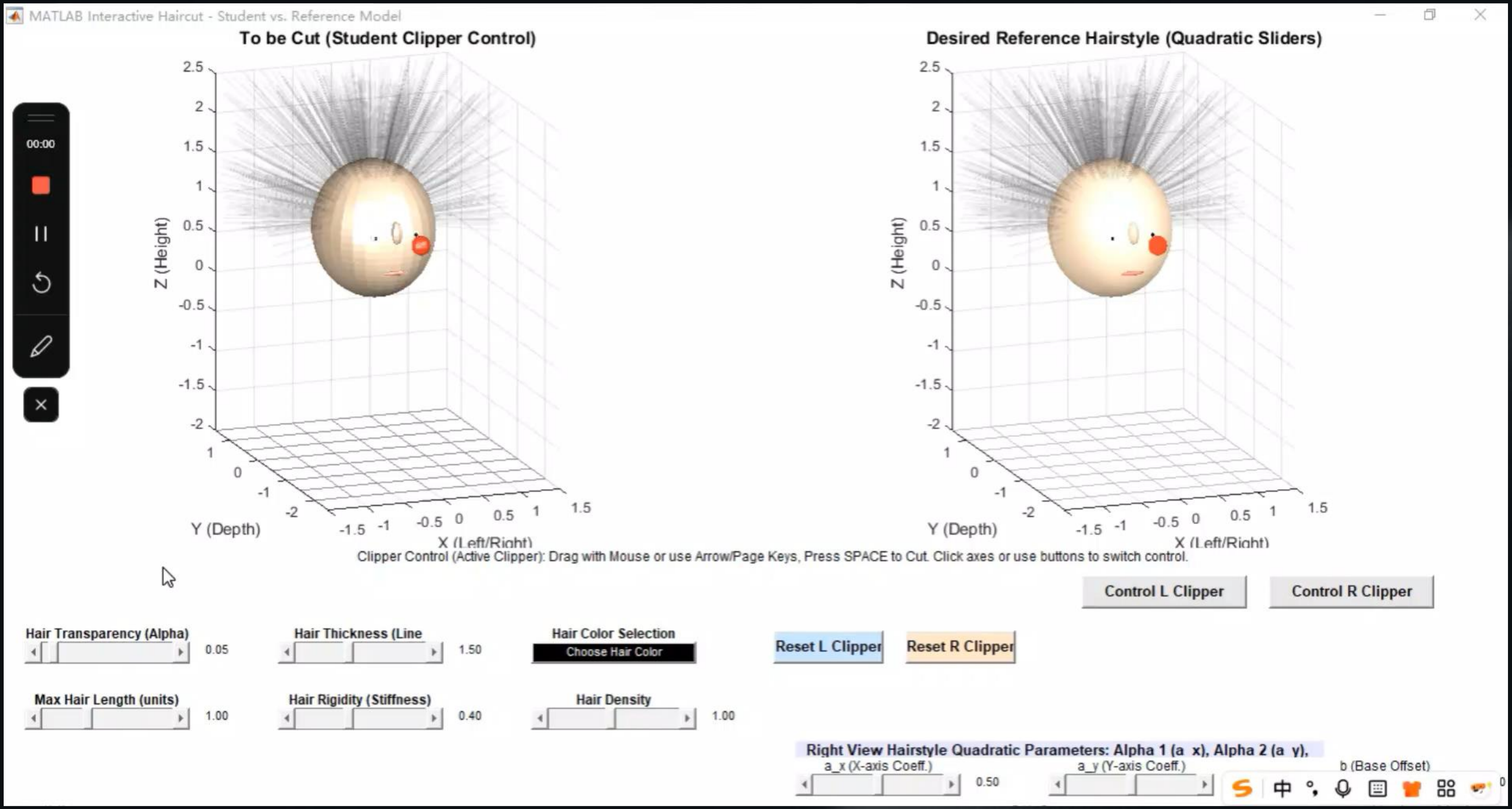
Technologies: Modular design, standardized processes, supply chain management.
Function: Reduce costs via modularization. Ensure consistency through quality control.
Challenges: Performance-cost balance, core component supply, production efficiency.

Business & Ecosystem

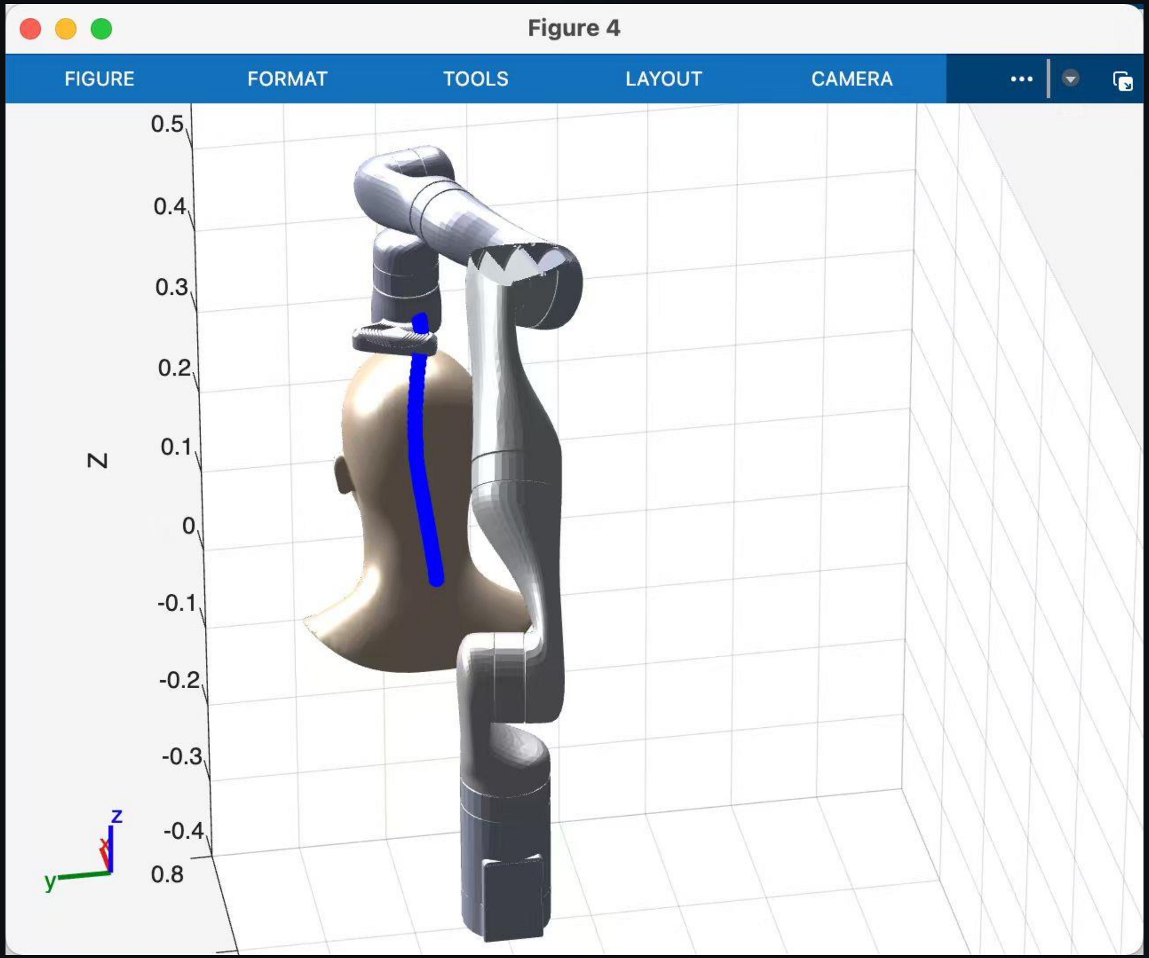
Technologies: IoT platforms, big data analytics, membership systems.
Function: Explore sales/leasing models. Build ecosystem via data analysis.
Challenges: Sustainable profit, user trust, industry standard formulation.

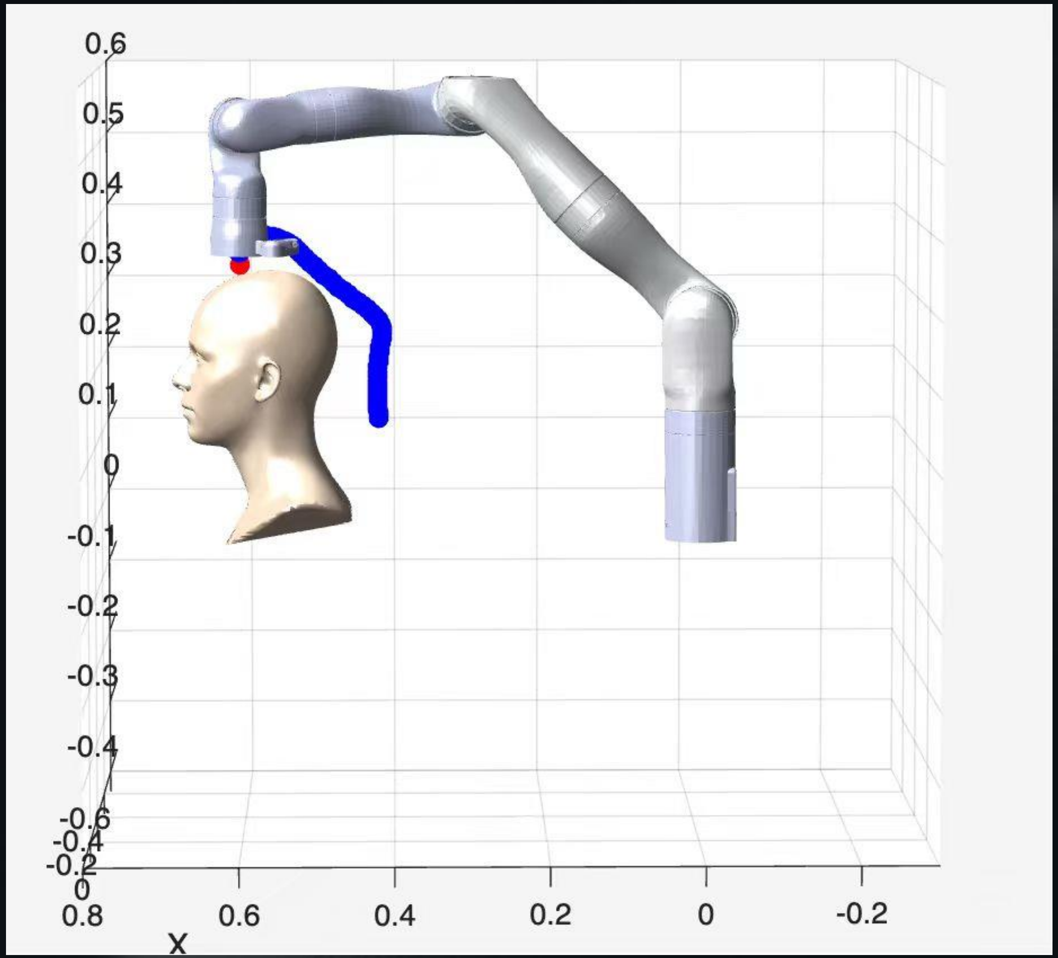
This layer serves as the bridge connecting technological innovation with market reality, ensuring scalability and profitability.

Preliminary Results: Hair Dynamics Simulator

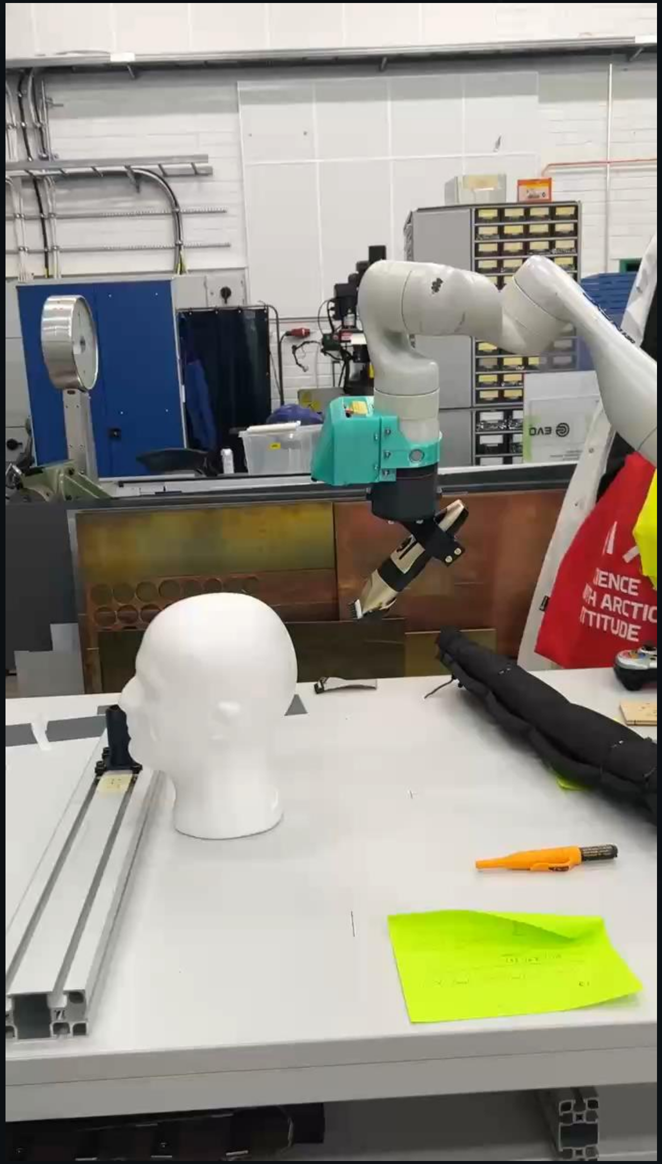


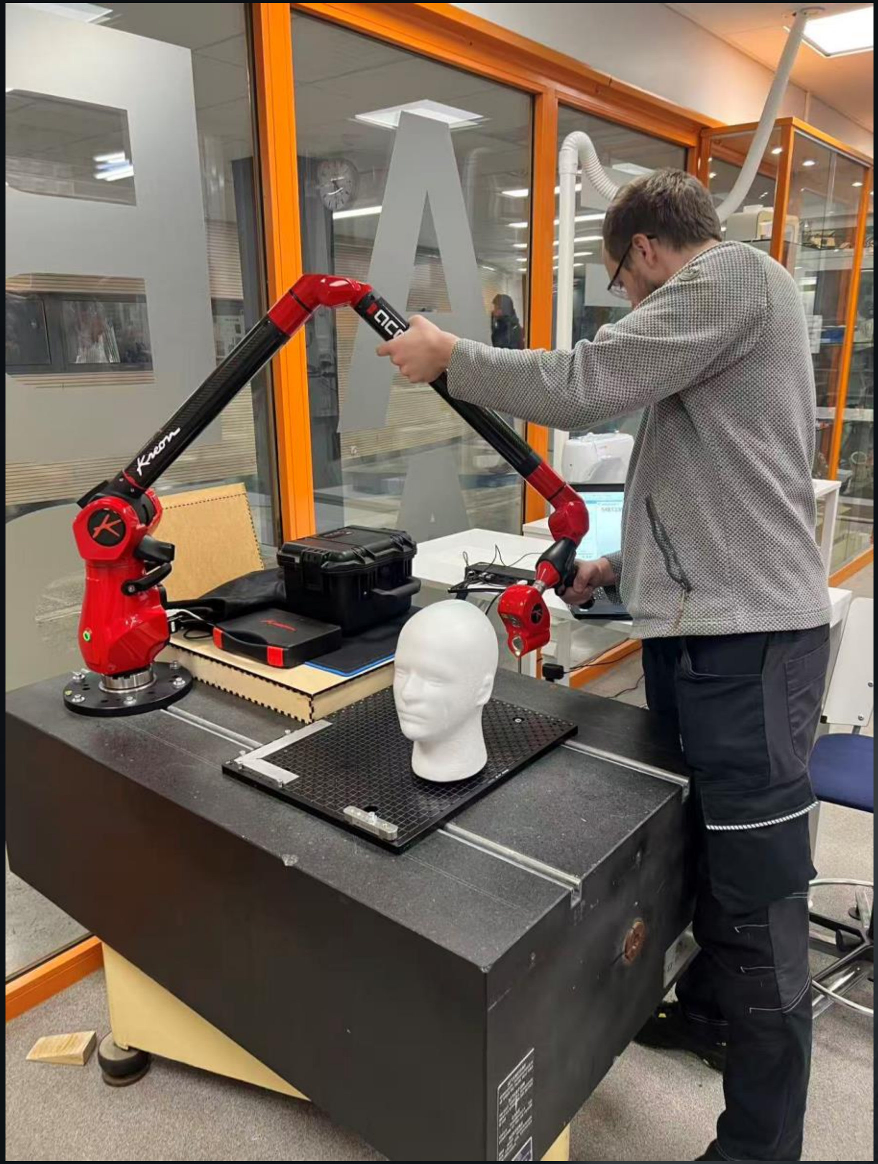
Preliminary Results: Haircutting Robot Simulator

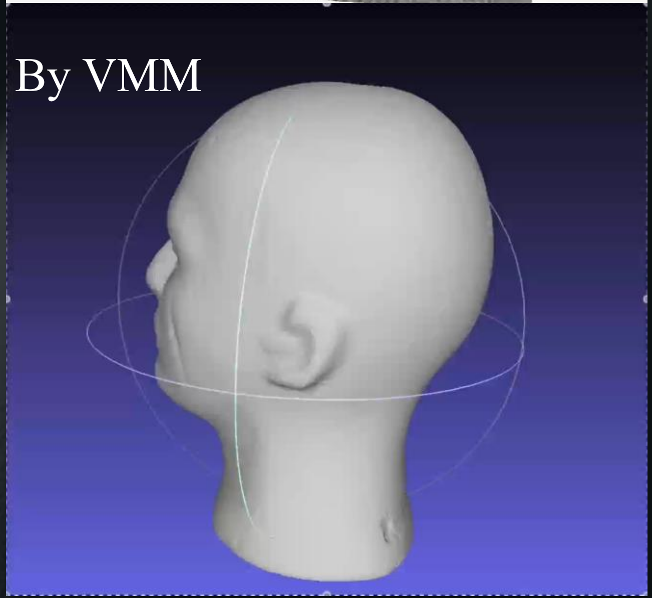




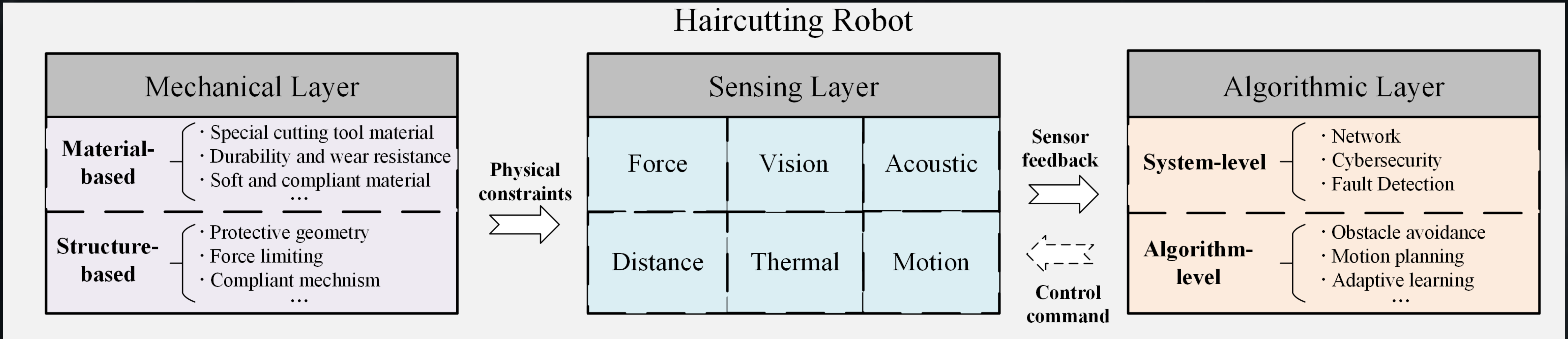
Preliminary Results: Experimental Systems

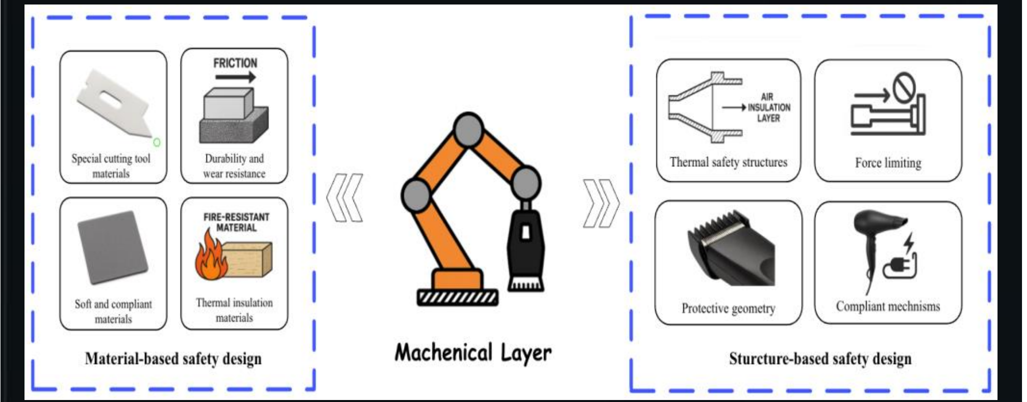


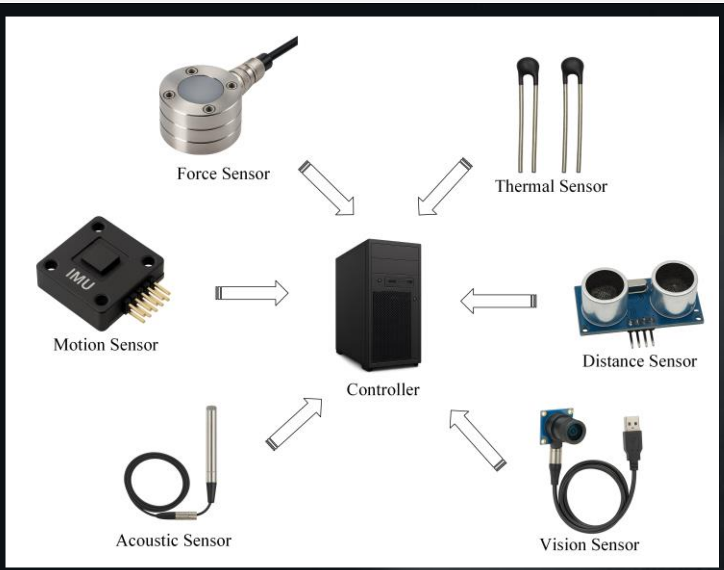


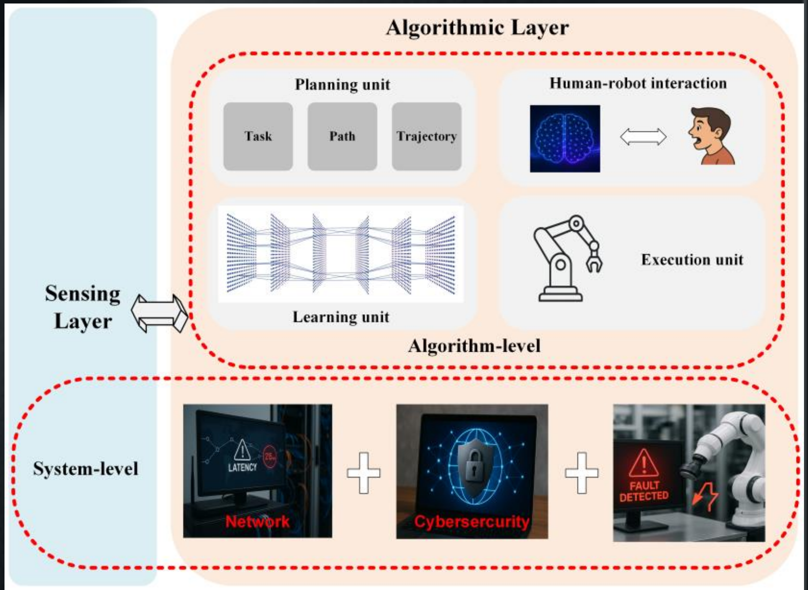


Preliminary Results: Safety Mechanisms

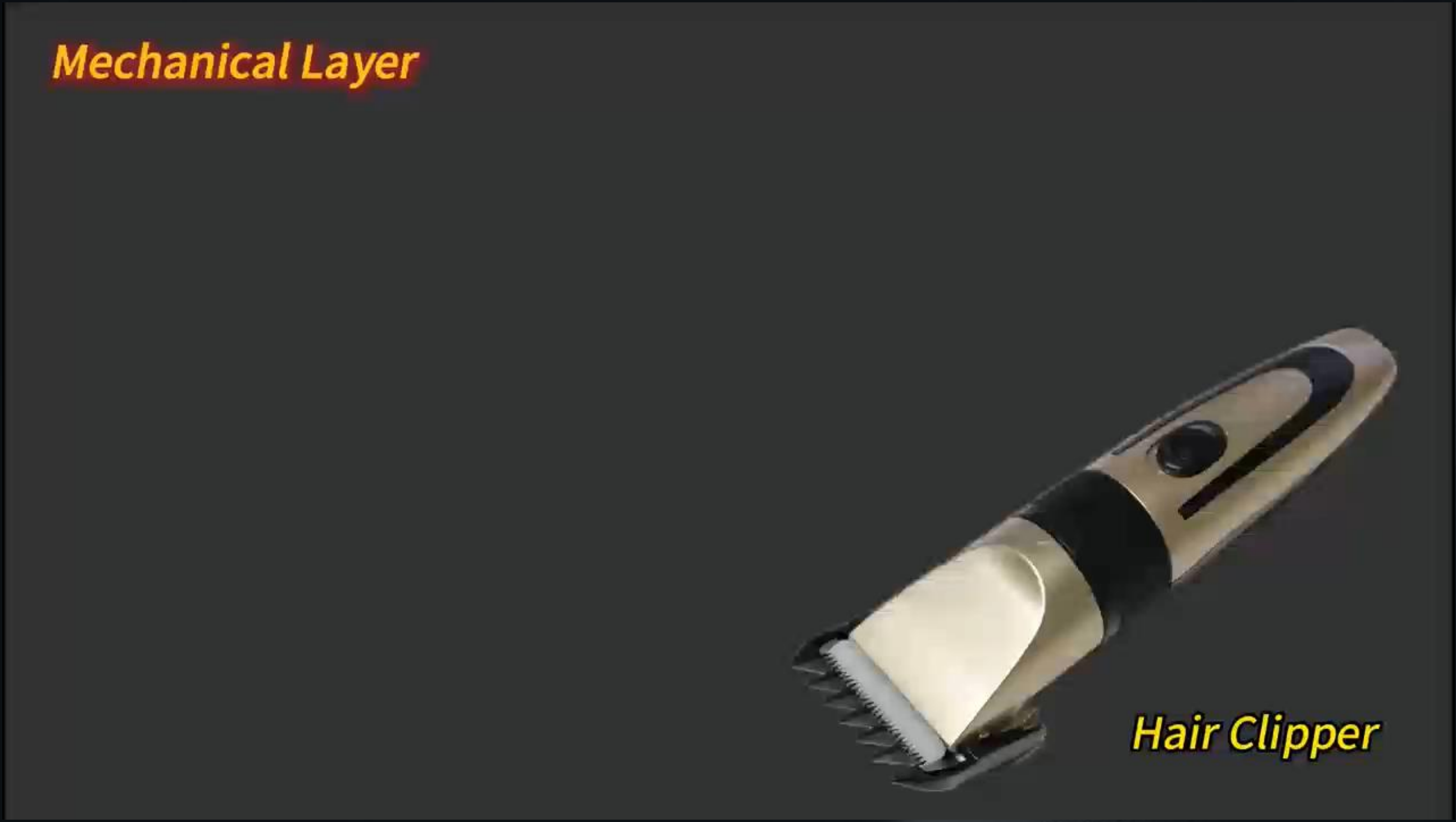






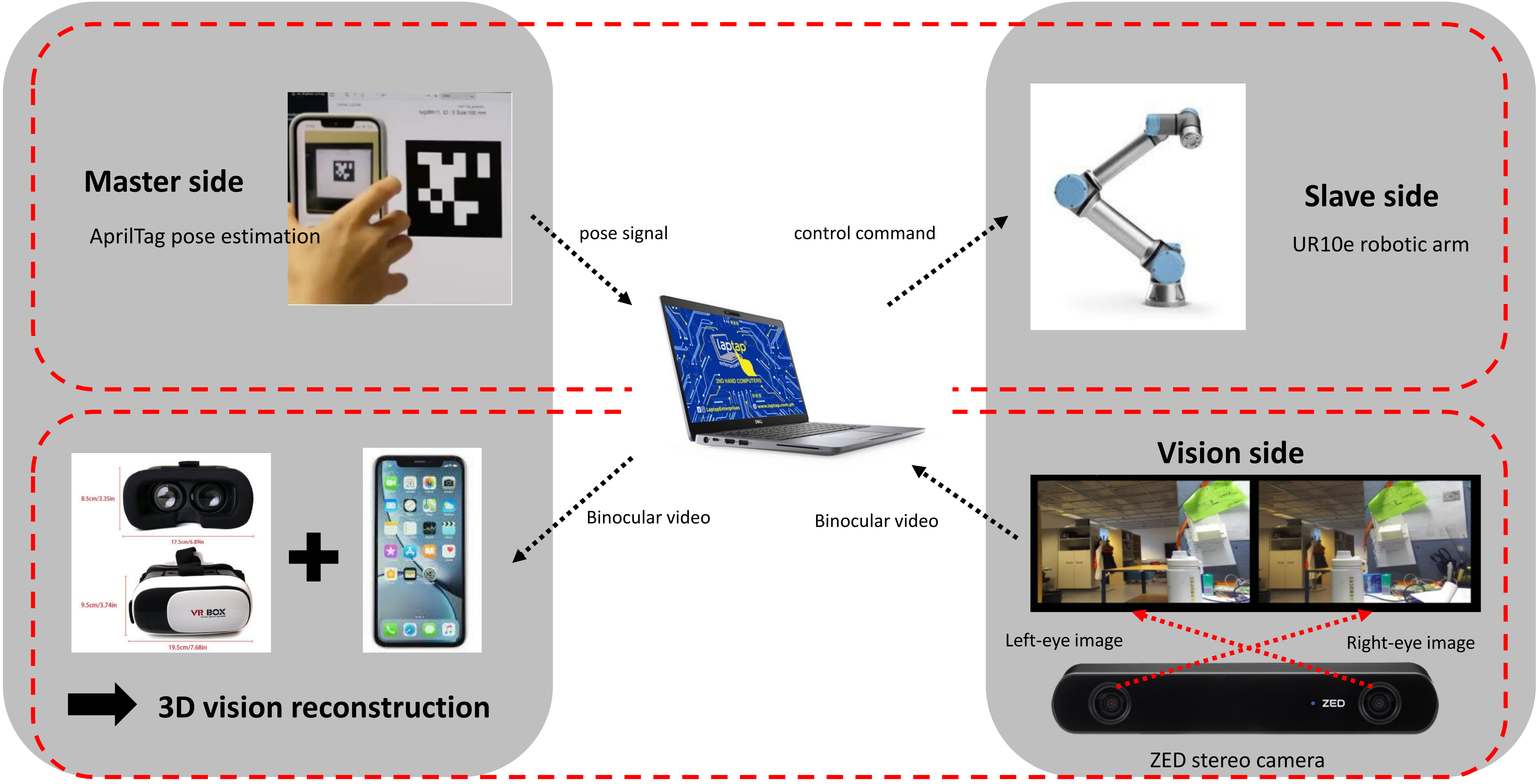


Preliminary Results: Safety Mechanisms

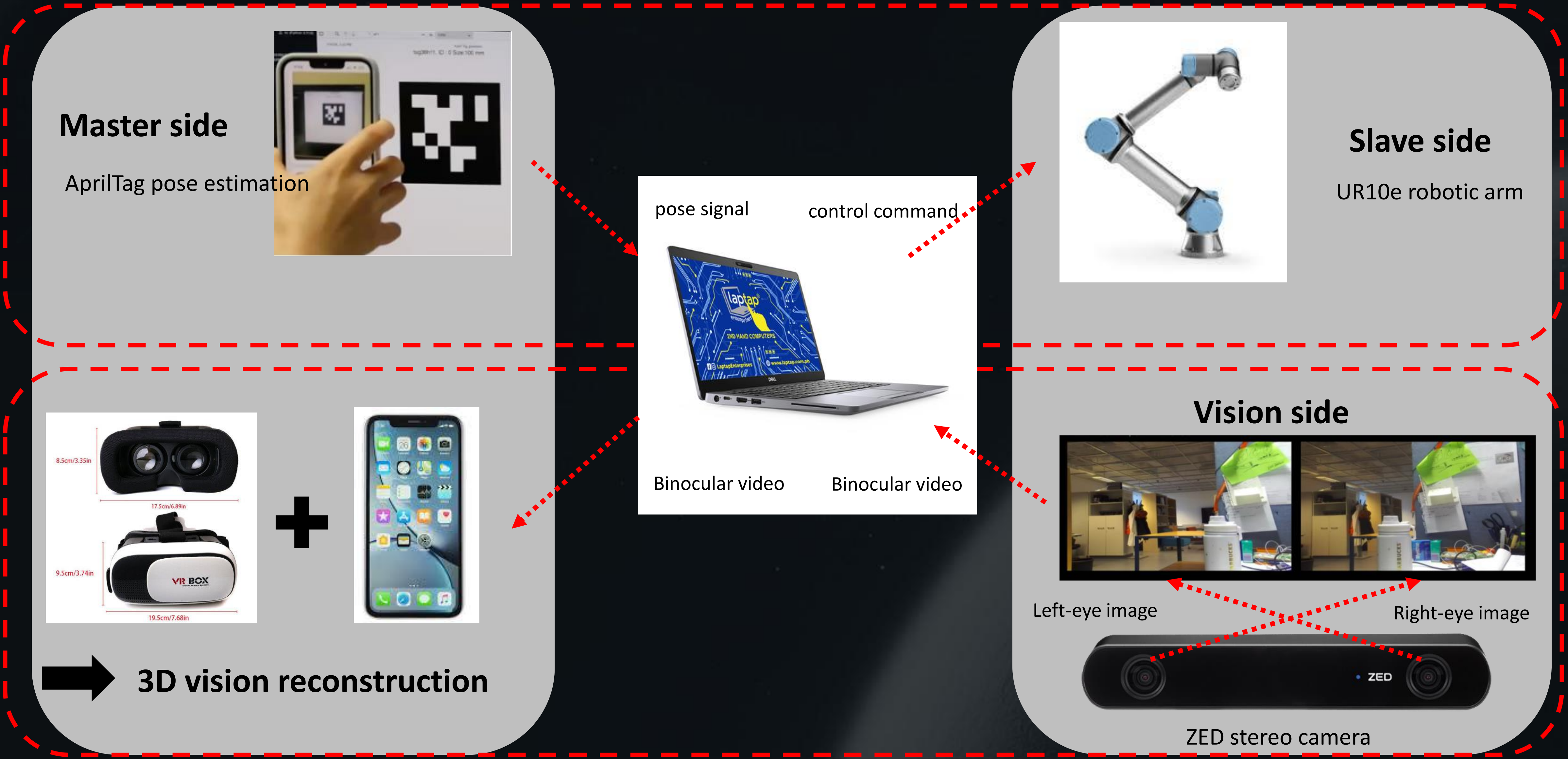


Z. Huang, A. Vilkki, S. Li and J. Röning, Safety in Robotic Haircutting, Under 2nd round of review in Journal of Field Robotics

Remote Teleoperation Robotic Haircutting System



Preliminary Results: Teleoperation



Preliminary Results: Teleoperation

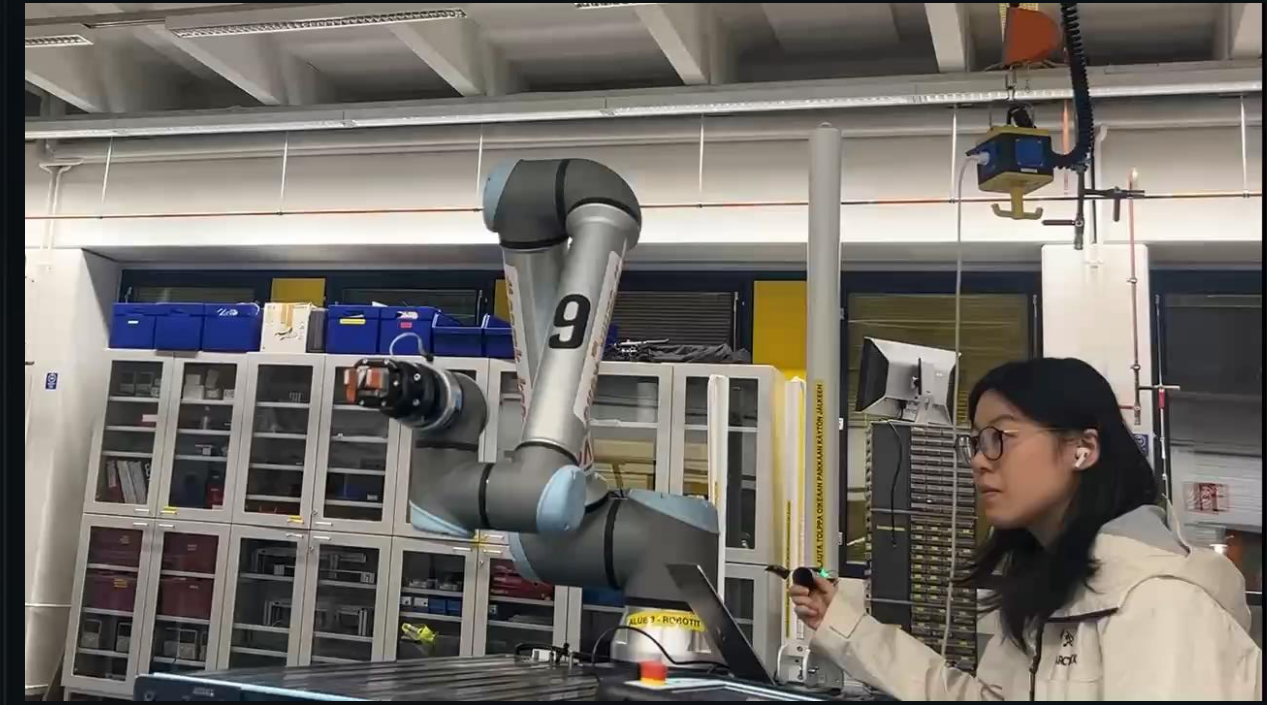
Lab-made wearable body tracking system for dual arm teleoperation

Lab-made wearable body tracking system for teleoperated haircutting

Lab-made wearable body tracking system

Preliminary Results: Teleoperation

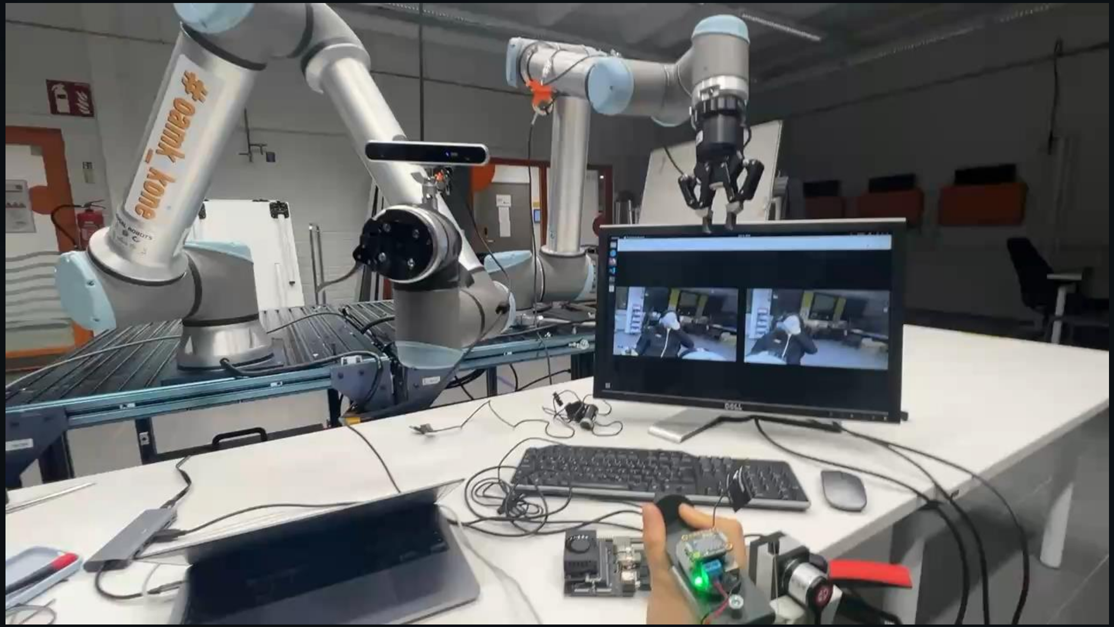
Lab-made control handle for teleoperated haircutting



Lab-made control handle (Cable CMM) for 6 DoF positioning (Ref. Z. Huang, Z. Chen, A. T. Khan, Y. Gan, F. Hu, J. Röning, and S. Li, A hybrid 6-DoF pose measurement system using string potentiometer and dual IMUs, 2025 IEEE International Workshop on Metrology for Sustainability, 2025)

Preliminary Results: Teleoperation

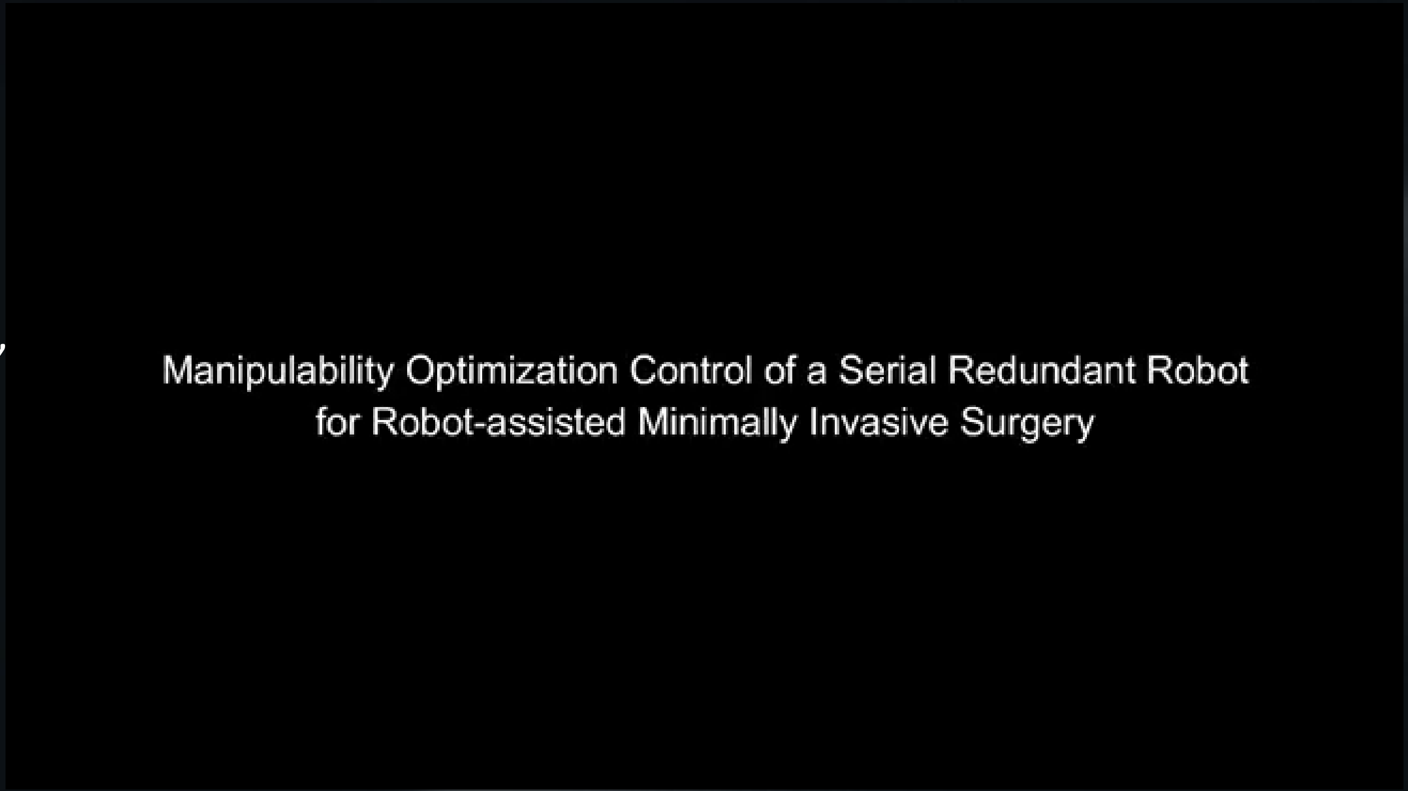
Experiments of Teleoperation in our lab.



Experiments of Teleoperation in our lab.

Preliminary Results: Teleoperation

Lab-made 6 DoF robotic arm using Xiaomi cybergear motors (Z. Chen, Z. Huang, Y. Gan, F. Hu, J. Röning, and S. Li, Development of a 6-DoF Robotic Manipulator with Smart Actuators and Real-Time Inverse Kinematics, ICNSC 2025.



Robotic surgery using self developed control algorithms (Ref. H. Su, S. Li, L. Bascetta, G. Ferrigno and Elena. Momi, Manipulability Optimization Control of a Serial Redundant Robot for Robot-assisted Minimally Invasive Surgery, ICRA 2019.)

07

Conclusion & Outlook

Key Takeaways

Convergence of Tech

Cheaper robots, rising labor costs, and advanced AI are creating a perfect storm for automation.

Practical Deployment

Supervised autonomy and robust safety engineering are the keys to early commercialization.

Massive Market

A multi-faceted digital platform approach unlocks a global market worth hundreds of billions.

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- Z. Chen, Z. Huang, Y. Gan, F. Hu, J. Röning, and S. Li, Development of a 6-DoF Robotic Manipulator with Smart Actuators and Real-Time Inverse Kinematics, ICNSC 2025
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Questions or Comments

Thanks!